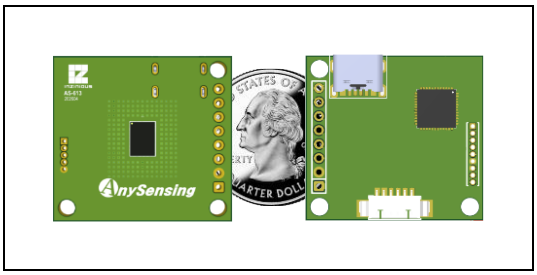
AS_613 with AI

Distance, Location, Human Vital-sign, AI Fall detection multi-channel FMCW Radar

with High performance, Small size, Low-power

GPIO, UART, USB, Indicate LED

User Manual



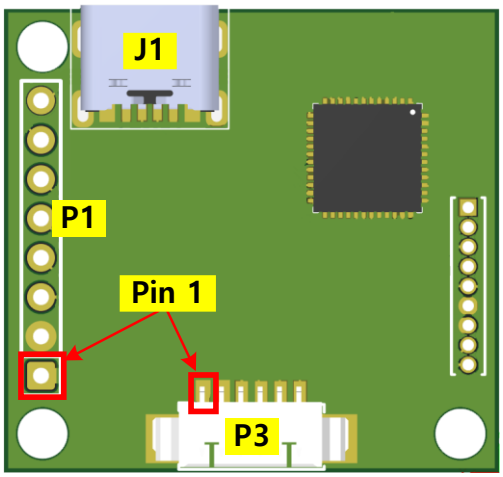
1 Features

- 60GHz multi-channel FMCW radar
- All-in-one radar and software solution
- Antenna and DSP software included
- Wide input voltage range: 4.5~5.3V
- Small size and easy to use
- GPIO, UART and USB interfaces
- Low power consumption: 350mW
- Multiple detection modes
  - Wall (side) mounting position measurement mode: max. distance 9m
  - Ceiling mounting position measurement mode: max. diameter 5m
  - Wall (side) mounting vital-sign measurement mode: max. distance 3.0m
  - Wall (side) mounting combined vital-sign and position mode: vital sign up to 3m, position up to 6m
  - Advanced vital-sign mode: includes HRV data such as RMSSD and SDNN (option)
  - Hospital ceiling-mounted patient monitoring mode (option)
  - Presence detection mode (option): wall or ceiling mounting; divides an 8m x 8m area into 256 zones (blocks) of 50cm and detects the presence of multiple people in real time
- Three power input options: USB-C, 2.54mm pin header, connector

2 Specification

Content		Definition
Model name		AS_613_AI
Hardware size		32.0 x 30.0 x 3.35 mm (No connector) 32.0 x 30.0 x 5.1 mm (Applying connector)
Weight		3.15 g
Transmit frequency		60GHz
Antenna		1 TX (EIRP : 10dBm), 3 RX / Antenna gain : 3dBi
RF pattern (-3dB width)		Horizontal 100° / Vertical 100°
Supply voltage / current		3.3 ~ 5.3 VDC / min. 210mA (If With AS_BW at least 410mA)
Operating / Storage temperature		-20 ~ 85 ℃ / -40 ~ 150 ℃
Interface		3 GPIO(3.3V level), 1 UART(3.3V level), USB
Function		Motion, Distance, Vital-sign(BR, HR, HRV)
Distance	Range	Distance mode : max. 9m (typ. 8m) Vital sign : max. 5m (typ. 3m)
	Resolution	Distance : 12 cm Vital sign : 85%

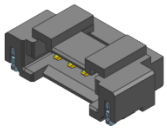
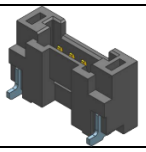
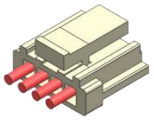
3 Pin description



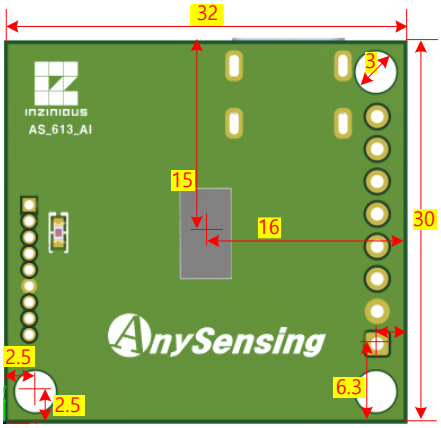
PIN	Name	P1	P3
1	VCC	Power + (4.5 ~ 5.3V)	Power + (4.5 ~ 5.3V)
2	GND	Power -	Power -
3	GPIO_user	Out(default) / in	Out(default) / in
4	NRST	N.C or High : Normal operation Low : Reset	N.C or High : Normal operation Low : Reset
5	UART TX	UART data transmit	UART data transmit
6	UART RX	UART data receive	UART data receive
7	GPIO_1	Out(default) / in	Not used
8	GPIO_2	Out(default) / in	Not used

P1 : 2.54mm Pin header
P3 : 12507WS-H06G (straight type) or 12507WR-H06G (90° type) .
J1 : USB-C connector

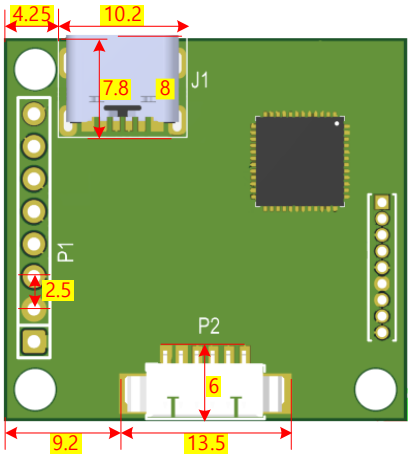
- AS_613_AI provides three GPIO pins (USER_IO, GPIO1, GPIO2); of these, the USER_IO pin outputs the target detection result in real time.
- When a target is detected, the USER_IO output is set High; when no target is detected, it is reset to Low. With the uiosec command, the output can be kept High for a specified time (0~999 s) after the target disappears.
- In the current firmware version, the GPIO1 and GPIO2 pins are not used for target detection output and always remain Low. Using the GPIO ports as inputs is not supported.

		
12507WR-H06G (Module side)	12507WS-H06G (Module side)	**12507HS-H06G1** (User side)

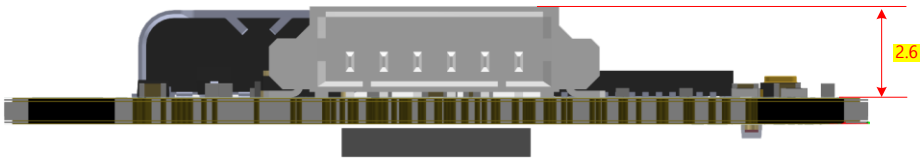
4 Dimension



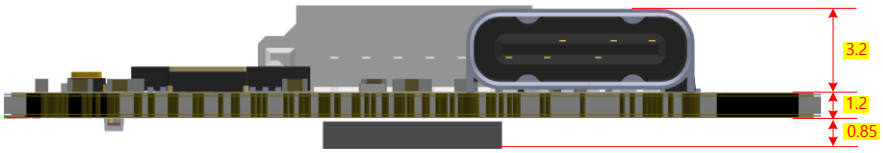
Top



Bottom



Install connector (12507WR-H06G)



Install USB-C connector

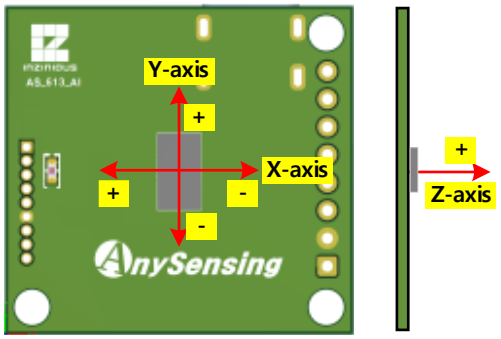
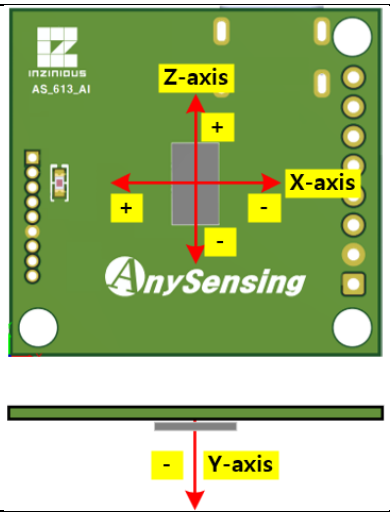
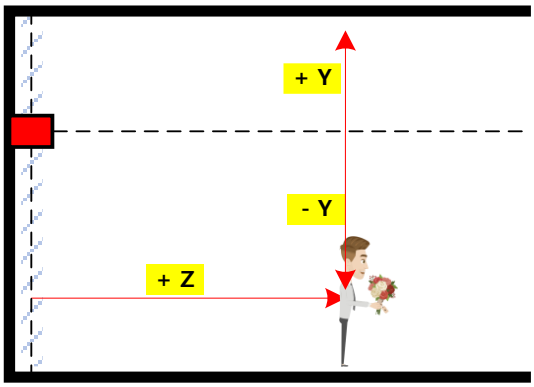
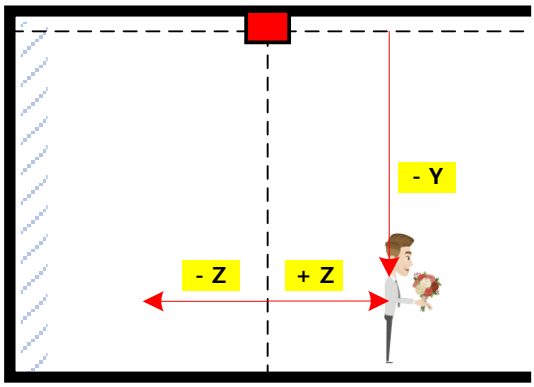
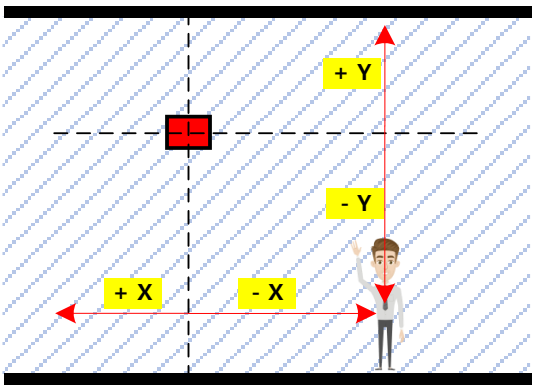
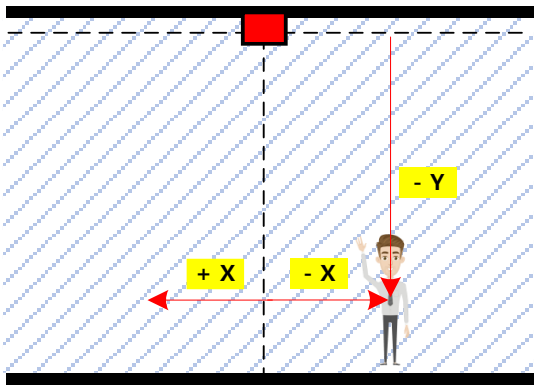
5 Function

Default Mode	Function
Wall mounting position measurement mode	**Mode 1:** Default mode: Mounted on a wall, it detects the position of people or moving objects. It provides the position of detected objects and additional information such as energy level. The basic mode outputs only the X and Z coordinates and the energy value of each object; the extended mode outputs the X, Y and Z coordinates and the energy value.
Ceiling mounting position measurement mode	**Mode 2:** Mounted on the ceiling, it detects people or moving objects on the floor. It outputs the position of detected objects as X and Z coordinates, or provides data on the detection area. The basic mode outputs only the X and Z coordinates and the energy value of each object; the extended mode outputs the X, Y and Z coordinates and the energy value.
Wall mounting presence (occupancy) detection mode	**Mode 3:** Mounted on a wall, it divides an 8m x 8m area into 256 zones (blocks) of 50cm and detects the presence (occupancy) of multiple people in real time. Unlike other modes that continuously track the coordinates of individual targets, it outputs a zone occupancy bitmap, an approximate head count and the duration.
Ceiling mounting presence (occupancy) detection mode	**Mode 4:** Mounted on the ceiling and looking down at the floor, it detects presence (occupancy). The decision method is the same as mode 3 (8m x 8m, 256 zones of 50cm); only the roles of the coordinate axes change to match the mounting orientation.
Wall or side mounting vital sign measurement mode	**Mode 5:** This mode detects the vital signs of a person within 2.5 meters and provides position, heart rate, respiration rate, heartbeat pulse, respiration waveform and motion data. The system supports a basic mode that outputs only vital-sign and distance information, and an extended mode that additionally includes position information.
Wall mounting vital sign and position measurement combined mode	**Mode 6:** A combined mode that detects vital signs and position at the same time. It detects position and heart rate up to 2.5m and respiration up to 3m, and position only up to 6m. Detects falls that occur in daily life and generates alerts.
Option Mode	
Advanced vital sign detection mode	**Mode 11:** The advanced vital-sign detection mode provides additional HRV data, including RMSSD and SDNN, in the extended mode of mode 5.
Restroom ceiling mounting mode	**Mode 12:** Mounted on a restroom ceiling, it outputs the presence status and respiration information of a person in the restroom. This mode can distinguish whether the person is standing, sitting or lying down.
Ceiling mounting hospital bed mode	**Mode 13:** Mounted on a hospital ceiling, it provides the patient's presence status on the bed, including posture, heart rate, respiration and HRV, distinguishes standing, sitting and lying states right beside the bed, and provides fall and pressure-ulcer risk alerts.

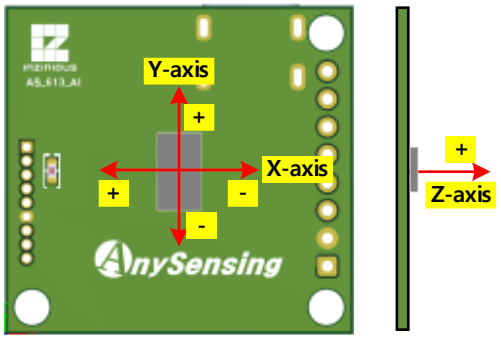
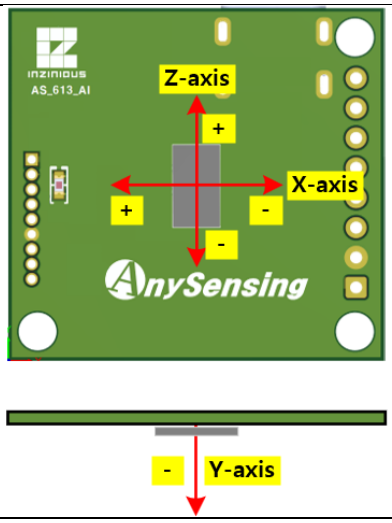
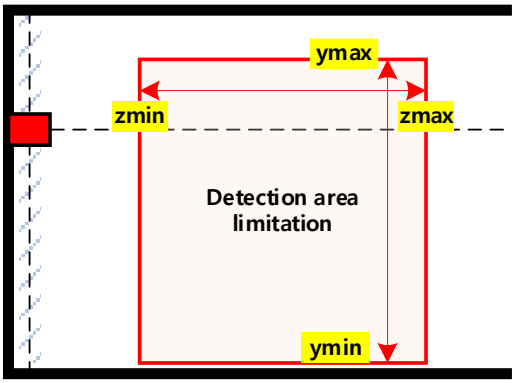
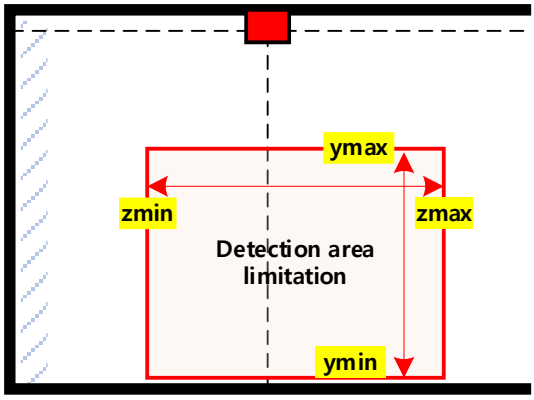
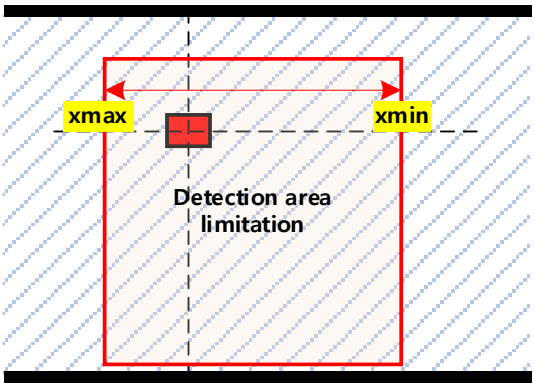
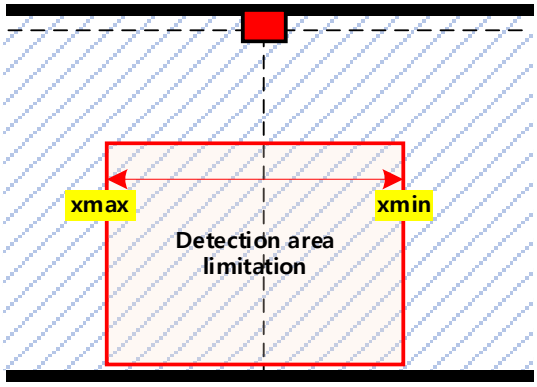
- If height information (Y-axis data) is needed, it can be calculated from the X-axis data, Z-axis data and linear distance data. The extended mode additionally includes Y-axis data and linear distance data.
- After the mode is changed, the first output data may not be correct because it is based on data from the previous mode.
- When entering pwr 0 (standby mode), the USER_IO output and LED turn off immediately regardless of the uiosec/ledsec hold-time settings (GPIO1 and GPIO2 are always Low and are not affected).
- The basic modes include all functions of the AS_613, and the option modes are enabled under a separate license agreement.
- Radar modules supplied as samples include the option modes,but for mass-production models this depends on the license status.

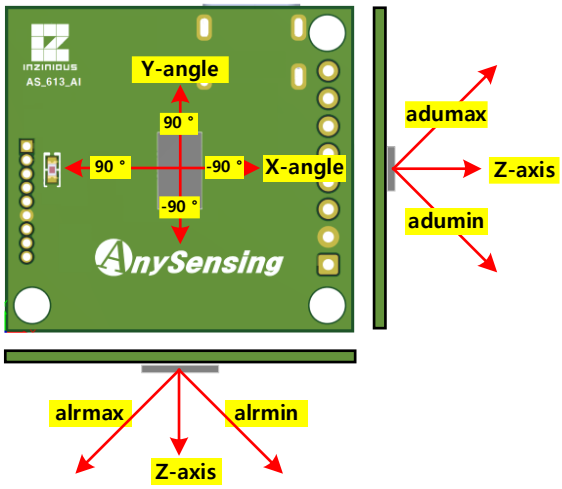
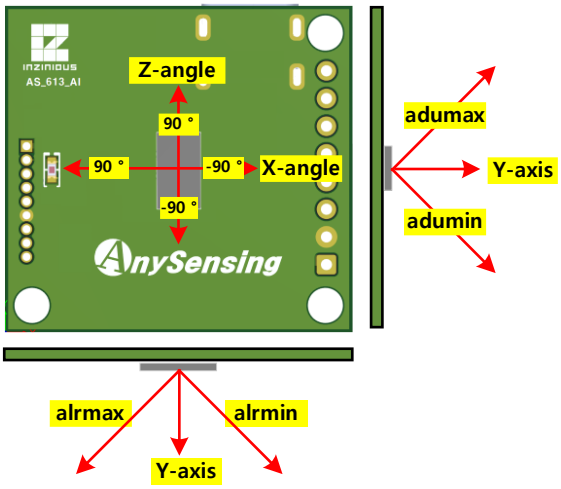
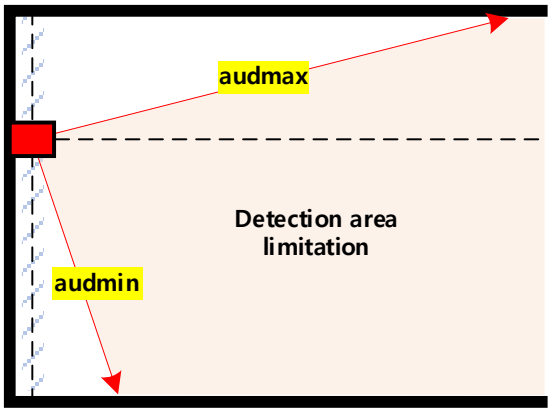
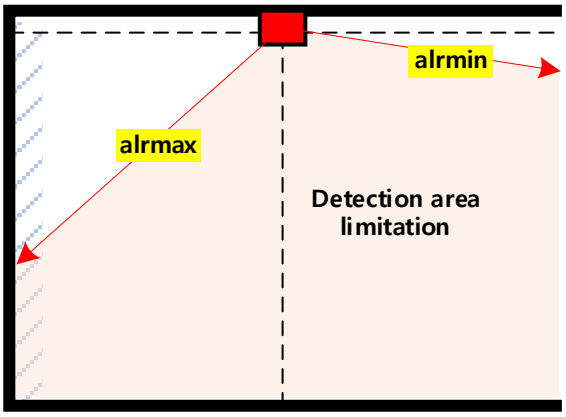
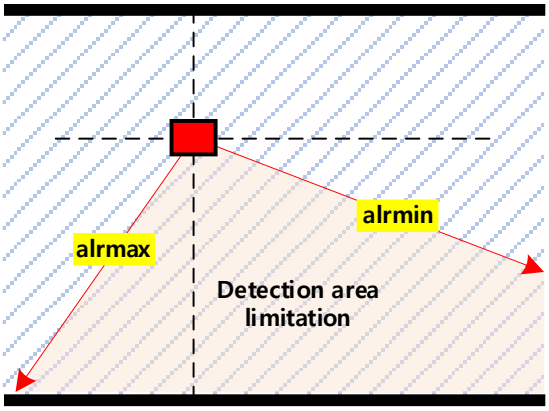
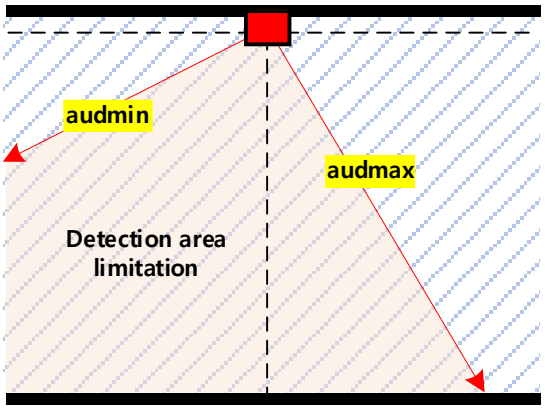
- **In vital-sign detection modes, the radar cannot single out only the person being monitored. If a moving object is closer than the person in the detection area, its motion cannot be separated from the person's, and the quality of the detected vital signs cannot be guaranteed. Movements the radar responds to include clock hands, air conditioners, fans, air purifiers, pets and the like. In particular, vital signs cannot be detected properly when the person's arms or hands are in front of the chest, such as when using a mouse, keyboard or smartphone, or when the arms are crossed.**

6 Radar coordinate orientation

<table>
<tr><th>Wall (side) installation</th><th>Ceiling installation</th></tr>
<tr><td> 
</td><td> 
</td></tr>
<tr><td> 
</td><td> 
</td></tr>
<tr><td> 
</td><td> 
</td></tr>
<tr><td>X-axis : (-) Left, (+) Right
Y-axis : (-) Down, (+) Up
Z-axis : (+) Front</td><td>X-axis : (-) Left, (+) Right
Y-axis : (-) Height
Z-axis : (-) Back, (+) Front</td></tr>
<tr><td colspan="2">Coordinates are output relative to the person, according to the wall or ceiling installation position.</td></tr>
</table>

Detection performance along the X axis is relatively better than along the Y or Z axis.

Wall (side) installation detection distance limit	Ceiling installation detection distance limit
	
	
	
Z-axis is always +	Y-axis is always -

Wall (side) installation detection angle limit	Ceiling installation detection angle limit
	
	
	

- Angles are set relative to the radar; the configurable range is -90° ~ 90° (factory defaults differ by mode and are mostly -75° ~ 75°), **and the stable detection range is -50° ~ 50°**.
- The detection area can be limited with distance settings and angle settings.
- Range settings provide a general limit on the range. Sensitivity to the angle settings in particular is

not high, so this function is approximate.

7 Mode Description

7.1 Wall mounting position measurement mode (mode 1)

- Mounted on the side of, or on a wall facing, the people or objects to be detected.

- The default detection distance settings are as follows:

x_min	-450 cm	x_max	450 cm
y_min	-100 cm	y_max	100 cm
z_min	40 cm	z_max	900 cm
alrmin	-75 °	alrmax	75 °
adumin	-75 °	adumax	75 °
Z is always positive (+).			

- The default detection angle is -75° to 75° on the X axis and -75° to 75° on the Y axis. The smaller the object, the narrower the detection angle. **The stable detection angle is -50° to 50°.**

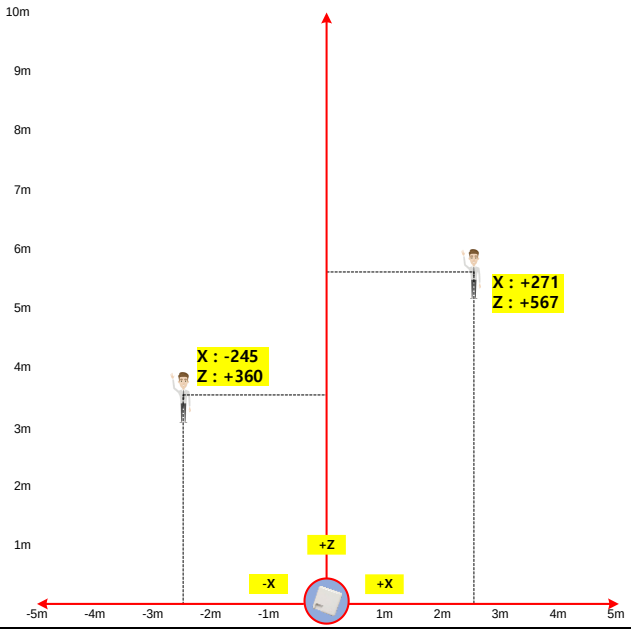
- The default number of detected objects is 3; up to 5 can be detected.
- By default, data is output 10 times per second; modes that output once per second or at a set time interval are also provided.

- **Detection may fail or accuracy may be reduced in the following cases:**

- Depending on the installation environment and the type of target, the actual detection distance may be shorter than the configured detection distance.
- A completely still person is not detected; a minimum amount of movement is required. Minimum movement includes breathing within 2m, talking within 5m, typing within 7m, and slow walking or shoulder movement beyond 7m.
- Normal detection is not guaranteed in environments with densely arranged furniture or equipment. Detection may fail or results may be inaccurate; in particular, in spaces with many metal objects or monitors, reflected waves may produce ghost objects. The best detection environment is an open space such as a meeting room.
- All moving objects in the detection space are detected. In particular, running fans in devices such as air conditioners, fans and air purifiers, curtains or long hanging cables may be detected, and moving pets are also detected. The sensor cannot distinguish between people, animals and objects, or measure their size.
- Even if a detected object does not move, the measured value may keep fluctuating within a range of 20~30cm.
- An object directly behind another object is not detected.
- People close to each other are detected as a single object. The degree depends on the distance from the radar; the farther from the radar, the larger the gap needed for them to be recognized as two objects.
- Because the maximum detection distance and range vary with the environment and angle of the space where the radar is installed, the maximum detection distance and range must be checked and tested in the actual environment.

- Data output format: ASCII

■ Wall mounting distance output mode



Basic data mode		
#ABBBBCCCDDD! (13 byte) (default mode)		
Item	**Byte**	**Content**
#	1	Data start character of the wall mounting distance output mode
A	1	Number identifying the detected object. e.g. 1 - first object, 2 - second object.
BBBB	4	X coordinate: left/right position relative to the radar, in cm. The first byte is the sign: '-' means left, '+' means right.
CCC	3	Z coordinate: front/back position relative to the radar, in cm.
DDD	3	Energy level of the object (0 ~ 999)
!	1	End-of-data character.
#0!		Output at the set time interval when no object is detected.
Extended data mode		
#ABBBBCCCCDDDEEE! (17 byte)		
Item	**Byte**	**Content**
#	1	Data start character of the wall mounting distance output mode
A	1	Number identifying the detected object. e.g. 1 - first object, 2 - second object.
BBBB	4	X coordinate: left/right position relative to the radar, in cm. The first byte is the sign: '-' means left, '+' means right.
CCCC	4	Y coordinate: down/up position relative to the radar, in cm. The first byte is the sign: '-' means down, '+' means up.
DDD	3	Z coordinate: front/back position relative to the radar, in cm.
EEE	3	Energy level of the object (0 ~ 999)
!	1	End-of-data character.
#0!		Output at the set time interval when no object is detected.

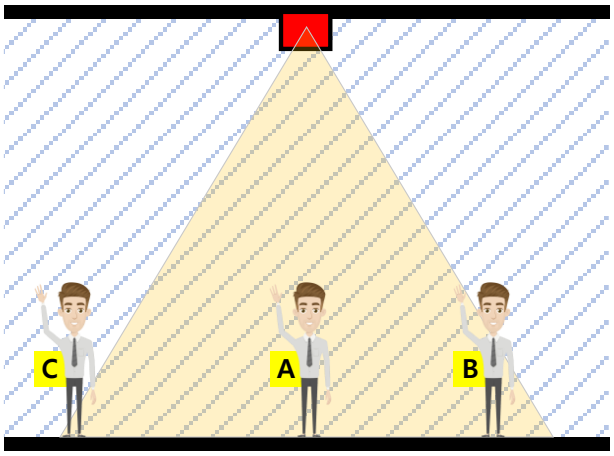
- No sign is shown because only positive distances in the Z direction are output.

7.2 Ceiling mounting position measurement mode (mode 2)

- Mounted on the ceiling, it detects the position of people or objects on the floor.
- The default detection distance settings are as follows:

x_min	-450 cm	x_max	450 cm
y_min	-450 cm	y_max	-40 cm
z_min	-450 cm	z_max	450 cm
alrmin	-80 °	alrmax	80 °
adumin	-80 °	adumax	80 °
Y always has - values			

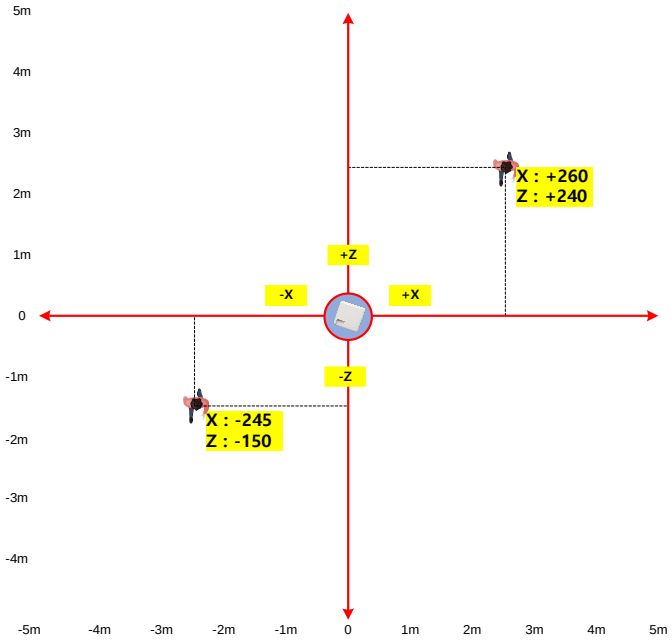
- The smaller the object, the narrower the detectable angle. **The stable detection angle is -50° to 50°.**
- Compared with wall mounting, the detection range is shorter. The higher the ceiling where the radar is installed, the wider the detection range. With a 2.5m ceiling, people are detected within a diameter of about 4~5m.
- **Detection may be impossible or accuracy may be reduced in the following cases:**



- Depending on the installation environment and the type of target, the actual detection distance may be shorter than the configured detection distance.
- In the figure above, A and B can be detected but C cannot, so the detection range is limited to B.
- Although A is closer to the radar than B, its reflection area is smaller than B's, so small breathing or movements may not be detected.
- Because the maximum detection range depends on the installation height and surroundings, the maximum detection distance must be confirmed by testing in the installation environment.
- When several people are close to each other or side by side as seen from the radar, they are recognized as a single object.
- The sensor may not work properly where furniture in the detection area is densely arranged, especially where there are many metal objects.

- Data output format: ASCII

■ Ceiling mounting distance output mode



Basic data mode

$ABBBBCCCCDDD! (14 byte) (default mode)

Item	Byte	Content
$	1	Data start character of the ceiling mounting distance output mode
A	1	Number identifying the detected object. e.g. 1 - first object, 2 - second object
BBBB	4	X coordinate: left/right position relative to the radar, in cm. The first byte is the sign: '-' means left, '+' means right.
CCCC	4	Z coordinate: front/back position relative to the radar, in cm. The first byte is the sign: '-' means back, '+' means front.
DDD	3	Energy level of the object (0 ~ 999)
!	1	End-of-data character.
$0!		Output at the set time interval when no object is detected.

Extended data mode

$ABBBBCCCCDDDDEEE! (18 byte)

Item	Byte	Content
$	1	Data start character of the ceiling mounting distance output mode
A	1	Number identifying the detected object. e.g. 1 - first object, 2 - second object
BBBB	4	X coordinate: left/right position relative to the radar, in cm. The first byte is the sign: '-' means left, '+' means right.
CCCC	4	Y coordinate: height between the object and the radar, with the radar facing the floor, in cm. The first byte is always the '-' sign.
DDDD	4	Z coordinate: front/back distance between the object and the radar, with the radar facing the floor, in cm. The first byte is the sign: '-' means back, '+' means front.
EEE	3	Energy level of the object (0 ~ 999)
!	1	End-of-data character.
$0!		Output at the set time interval when no object is detected.

7.3 Wall mounting presence (occupancy) detection mode (mode 3)

- Mounted on a wall from which the space to be monitored is visible. Unlike other modes that continuously track the coordinates of individual targets, this dedicated mode divides an 8m x 8m area (left-right and distance directions) into 256 zones (blocks) of 50cm, detects the presence (occupancy) of each zone in real time, and outputs the result (head count, duration and occupancy bitmap). It is suitable for multi-person presence detection.
- Presence (occupancy) detection mode does not give an exact head count. It indicates the approximate number of people, the distribution of occupancy and the approximate occupancy time.

- The default detection distance settings are as follows:

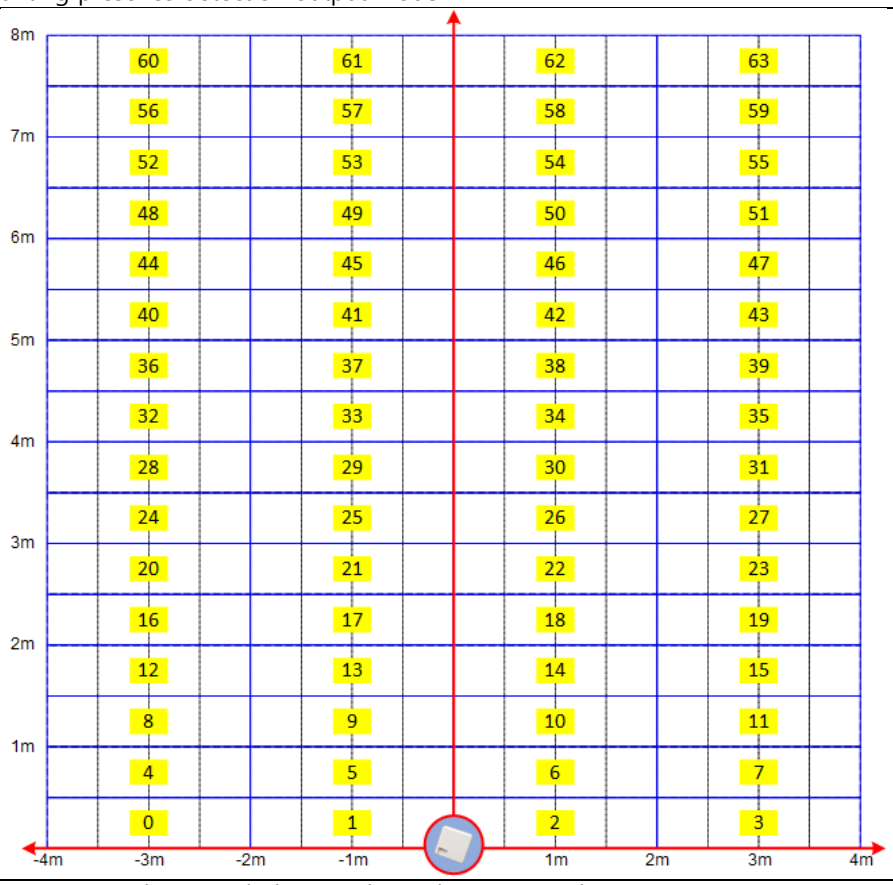
x_min	-400 cm	x_max	400 cm
y_min	-200 cm	y_max	200 cm
z_min	0 cm	z_max	800 cm
alrmin	-75 °	alrmax	75 °
adumin	-75 °	adumax	75 °
Z is always positive (+).			

- The left-right (X-axis) origin of the actual grid is fixed at -400cm, so even if the detection area is adjusted with xmin/xmax, coordinates detected outside it are assigned to and displayed in the nearest edge zone.
- A zone is confirmed as occupied when a detection occurs in at least 6 of the last 12 frames (about 0.6 s at 20 fps) and its position spread is within 30cm, or, even if the spread exceeds 30cm, when the movement path proceeds consistently in one direction (without going back and forth or deviating greatly). The senin command is stored with the same range as in other modes but is not used in this confirmation.
- The occupancy hold time of a zone is set with the senout command. In this mode, a value of 20~500 s must be entered in 10 s steps (e.g. senout 300); the default is 300 s (5 min). Values that are not multiples of 10 are rejected with =OverRange!. However, a zone whose detection evidence has stopped is cleared after half the senout value or 20 s, whichever is shorter, so values of 40 s or more behave identically.
- The LED and USER_IO pin turn on when at least one zone is confirmed as occupied, and turn off when all zones are empty. Even if a person stands still in a zone, they stay on as long as that zone remains occupied.

- **Detection may be impossible or accuracy may be reduced in the following cases:**
- If people stand in a row across several zones within 71cm of each other, they may be merged and counted as fewer people than are actually present.
- If a moving object (fan, curtain, etc.) is closer to the sensor than the target, its zone may be falsely marked as occupied. This can be solved by registering that location as a detection exclusion zone with exzon/exzset.
- The wave, hmax and bmax commands return =UnSupport! in this mode. The exten and interval commands store their values but do not affect the output of this mode. The exzpick command returns =UnSupport! because this mode does not use the coordinate tracker; exclusion zones must be registered directly with exzset. To exclude by zone (block) number, use the pblkset command.
- Immediately after entering the mode, and immediately after switching between mode 3 and mode 4, occupancy is re-evaluated from scratch. Even if people are present, they are shown as occupied only after a re-confirmation of at least 12 frames (about 0.6 s), and the first output packet is sent 2 s after entry with the state at that moment.
- Presence output packets are sent according to state changes, not at a fixed period. A packet is sent immediately when occupancy appears in an empty room or when all zones become empty. While occupied, a packet is sent at intervals of up to 0.5 s when the zone occupancy bitmap changes, every 30 s when it does not change, and every 60 s while the room stays empty. Right after entering the mode, the sensor only calculates for 2 s and then sends the state at that moment once.

- Data output format: ASCII

■ Wall mounting presence detection output mode



*ABBBCCCC...CCCC! (70 bytes excluding newline, when occupied)

Item	Byte	Content
*	1	Data start character.
A	1	Number of people present (HEX, 0 ~ F). Fixed at the maximum if exceeded
BBB	3	Duration (in minutes, HEX, 000 ~ FFF) for which at least one person has occupied any zone. Fixed at the maximum if exceeded
CCCC...CCCC	64	Occupancy bitmap of 256 zones (16 x 16, 50cm spacing) (HEX, 64 digits). Starting from the first character on the left, each bit corresponds to zone 0, zone 1 and so on; 1 means the zone is occupied.
!	1	End-of-data character.
*0!	*0!	Short packet sent when all 256 zones are empty (vacant). The head-count, duration and bitmap fields are omitted.

Even if the detection area is limited and the area sensed by the sensor becomes smaller, the data output format stays the same.

7.4 Ceiling mounting presence (occupancy) detection mode (mode 4)

- Mounted on the ceiling and looking down at the floor, this mode detects presence (occupancy). The decision method is the same as mode 3 (wall mounting) (8m x 8m, 256 zones of 50cm); only the roles of the coordinate axes change to match the mounting orientation — the X axis is left-right, the Z axis is the other horizontal axis, and the Y axis is the height from the ceiling toward the floor (downward).
- Presence (occupancy) detection mode does not give an exact head count. It indicates the approximate number of people, the distribution of occupancy and the approximate occupancy time.

- The default detection distance settings are as follows:

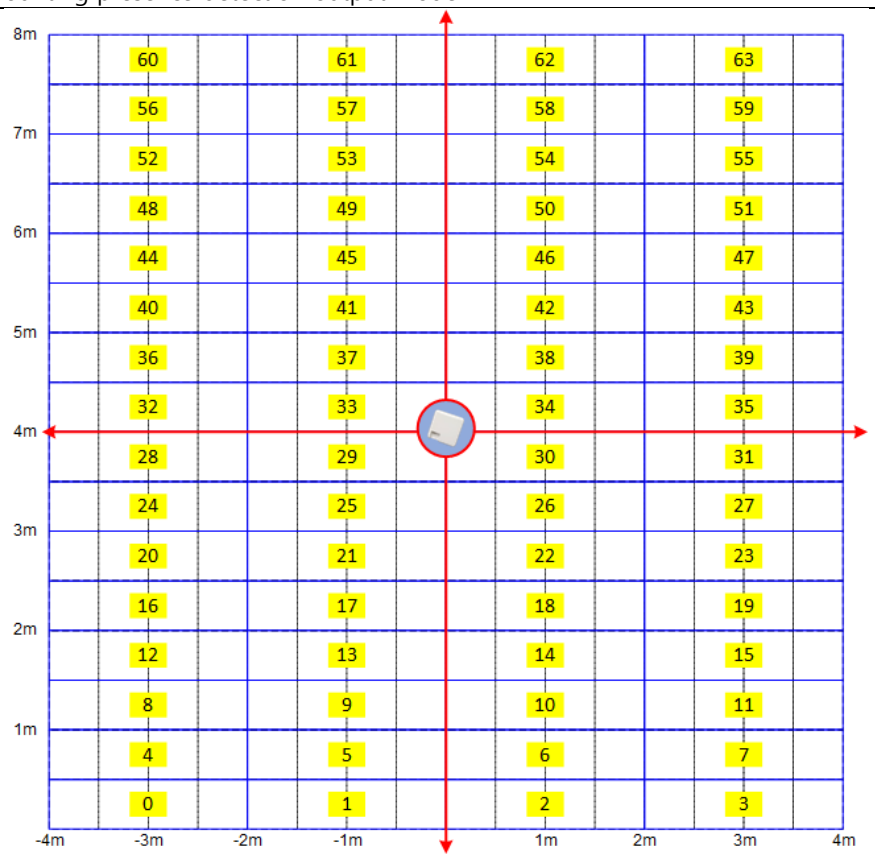
x_min	-400 cm	x_max	400 cm
y_min	-300 cm	y_max	-30 cm
z_min	-400 cm	z_max	400 cm
alrmin	-75 °	alrmax	75 °
adumin	-75 °	adumax	75 °
Y is always negative (-).			

- In ceiling mounting modes, the xmin/xmax/zmin/zmax commands accept a range of ±450cm. The factory default detection area (±400cm on each of the X and Z axes) is within this range, so values can be freely narrowed and widened again by command. To restore the factory defaults (±400cm) exactly, use mreset or freset.
- Due to ceiling mounting, when two different detections, such as the torso and legs of one person, overlap within 30cm in floor projection, the farther detection is automatically suppressed and not reflected in the grid (unless the nearer reflection is weaker by 10dB or more). This function is always on and cannot be turned off with the gnd command. If two people stand within 30cm of each other, one of them may be missed in the occupancy decision.
- The zone occupancy confirmation/hold time (senout, default 300 s / 20~500 s in 10 s steps) and LED/USER_IO behavior (on when any zone is confirmed as occupied) are the same as in mode 3.
- The gnd command can be set and queried in this mode (common to ceiling mounting modes), but since this mode does not use the coordinate tracker, it does not affect the actual presence decision.

- **Detection may be impossible or accuracy may be reduced in the following cases:**
- If two people (or the torso/legs of one person) overlap within 30cm in floor projection, a person may be missed due to the suppression function described above.
- If people stand in a row across several zones within 71cm of each other, they may be merged and counted as fewer people than are actually present.
- The wave, hmax and bmax commands return =UnSupport! in this mode. The exten, interval and gnd commands store their values but do not affect the presence decision of this mode. The exzpick command returns =UnSupport! because this mode does not use the coordinate tracker; exclusion zones must be registered directly with exzset. To exclude by zone (block) number, use the pblkset command.
- Immediately after entering the mode, and immediately after switching between mode 3 and mode 4, occupancy is re-evaluated from scratch, and the first output packet is sent 2 s after entry with the state at that moment. While occupied, packets for a changed zone occupancy bitmap are sent at intervals of up to 10 s, unlike mode 3 (0.5 s).

- Data output format: ASCII

■ Ceiling mounting presence detection output mode



~ABBBCCCC...CCCC! (70 bytes excluding newline, when occupied)

Item	Byte	Content
~	1	Data start character.
A	1	Number of people present (HEX, 0 ~ F). Fixed at the maximum if exceeded
BBB	3	Duration (in minutes, HEX, 000 ~ FFF) for which at least one person has occupied any zone. Fixed at the maximum if exceeded
CCCC...CCCC	64	Occupancy bitmap of 256 zones (16 x 16, 50cm spacing) (HEX, 64 digits). Starting from the first character on the left, each bit corresponds to zone 0, zone 1 and so on; 1 means the zone is occupied.
!	1	End-of-data character.
~0!	~0!	Short packet sent when all 256 zones are empty (vacant). The head-count, duration and bitmap fields are omitted.

7.5 Side wall mounting vital-signal measurement mode (mode 5)

- A dedicated mode for vital-sign detection with a maximum detection range of 2.5m.
- The default detection distance settings are as follows:

x_min	-125 cm	x_max	125 cm
y_min	-150 cm	y_max	100 cm
z_min	40 cm	z_max	250 cm
alrmin	-75 °	alrmax	75 °
adumin	-75 °	adumax	75 °
Z always has + values			

- **The stable detection angle is -50° to 50°.**
- Detects and outputs the vital signs of the person closest to the sensor.
- The data provided are respiration rate, heart rate, respiration waveform, heartbeat waveform and detection distance. HRV (RMSSD, SDNN) is provided in mode 11.
- In a stable state, the measured respiration rate has an accuracy of 85% or more, and the heart rate about 85%.
- This mode supports the extended data mode. The extended data mode is enabled with the 'exten 1₩n' command and disabled with the 'exten 0₩n' command.

- When the extended mode is enabled, the following additional data are provided:

Normal mode	Vital status Heart rate Breath rate Heart wave data Breath wave data Option data: distance
Extension mode	This mode includes all data of the normal mode and provides additional information. X-axis distance information Y-axis distance information Z-axis distance information Straight-line distance information

- **Detection may be impossible or accuracy may be reduced in the following cases:**

- The sensor measures vital signs by detecting the frequency of the small back-and-forth chest movements while a person breathes or the heart beats.
- The radar detects vital signs from the small chest movements that occur when breathing or when the heart moves. If the person intentionally breathes rapidly or holds their breath, the chest is compressed and vital signs are not detected properly. Under these conditions in particular, the heartbeat may not be detected.
- The best detection performance is achieved in a stable state. A stable state means the person is not moving or talking, with hands down and chest facing the radar sensor.
- If the arms are crossed, the hands are raised in front of the chest at a desk (e.g. when typing), or the hands cover the chest while moving, detection may fail or abnormal values may be output.
- Normal data cannot be output if there is a moving object between the sensor and the target, or if an object is closer than the target.
- Vital signs can be detected even when the sensor faces the person's back or arm, but accuracy is reduced.
- If there are moving objects such as fans or robot vacuum cleaners in the detection area, detection signals may be output.
- Use of the sensor is not recommended in surroundings with complex objects.

- Data output format: ASCII

■ Mode without vital-sign waveform: default mode

Basic data mode		
%ABBBCCDDD! (11 byte) (default mode)		
Item	**Byte**	**Content**
%	1	Start character indicating data output without vital-sign waveform
A	1	Vital-sign detection status 0: No target 1: Target present, moving 2: Target present, no vital signs 3: Target present, vital signs detected
BBB	3	Heart rate per minute
CC	2	Respiration rate per minute
DDD	3	Detection distance (cm)
!	1	End-of-data character
%0!		Output at the set time interval when no object is detected
Extended data mode		
%ABBBCCDDDDEEEEFFFGGG! (22 byte)		
Item	**Byte**	**Content**
%	1	Start character indicating data output without vital-sign waveform
A	1	Vital-sign detection status 0: No target 1: Target present, moving 2: Target present, no vital signs 3: Target present, vital signs detected
BBB	3	Heart rate per minute
CC	2	Respiration rate per minute
DDDD	4	X coordinate: left/right position relative to the radar, in cm. The first byte is the sign: '-' means left, '+' means right.
EEEE	4	Y coordinate: down/up position relative to the radar, in cm. The first byte is the sign: '-' means down, '+' means up.
FFF	3	Z coordinate: front position relative to the radar, in cm.
GGG	3	Detection distance (cm)
!	1	End-of-data character
%0!		Output at the set time interval when no object is detected

- The initial state of this mode is the basic data mode without vital-sign waveform data. To include vital-sign waveform data, use 'wave 1₩n', and to output extended data, use 'exten 1₩n' for additional settings.

■ Mode with vital-sign waveform: output 10 times per second

Basic data mode		
@ABBBCCDEEEFFF! (15 byte)		
Item	**Byte**	**Content**
@	1	Start character indicating data output with vital-sign waveform
A	1	Vital-sign detection status 0: No target 1: Target present, moving 2: Target present, no vital signs 3: Target present, vital signs detected
BBB	3	Heart rate per minute
CC	2	Respiration rate per minute
D	1	Heartbeat pulse, 0, 1
EEE	3	Respiration waveform data, 0 ~ 999
FFF	3	Detection distance (cm)
!	1	End-of-data character
@0!		Output at the set time interval when no object is detected
Extended data mode		
@ABBBCCDEEEFFFFGGGGHHHIII! (26 byte)		
Item	**Byte**	**Content**
@	1	Start character indicating data output with vital-sign waveform
A	1	Vital-sign detection status 0: No target 1: Target present, moving 2: Target present, no vital signs 3: Target present, vital signs detected
BBB	3	Heart rate per minute
CC	2	Respiration rate per minute
D	1	Heartbeat pulse, 0, 1
EEE	3	Respiration waveform data, 0 ~ 999
FFFF	4	X coordinate: left/right position relative to the radar, in cm. The first byte is the sign: '-' means left, '+' means right.
GGGG	4	Y coordinate: down/up position relative to the radar, in cm. The first byte is the sign: '-' means down, '+' means up.
HHH	3	Z coordinate: front position relative to the radar, in cm.
III	3	Detection distance (cm)
!	1	End-of-data character
@0!		Output at the set time interval when no object is detected

7.6 Wall mounting vital-signal and position measurement combined mode (mode 6)

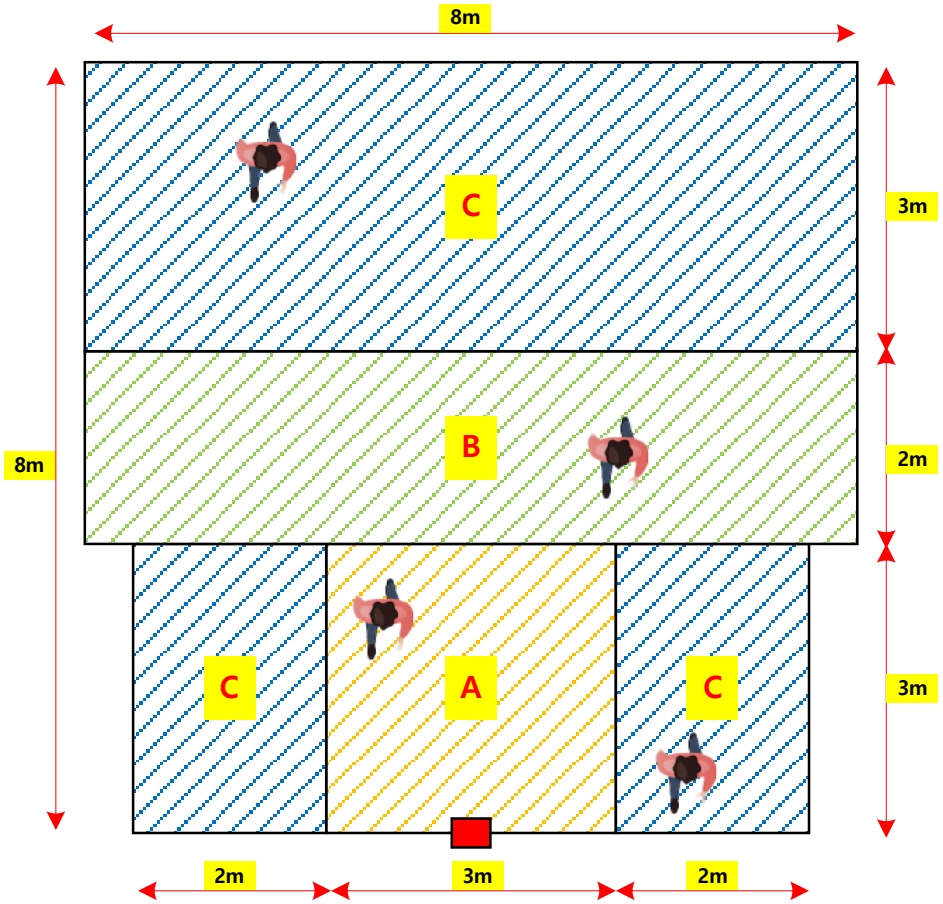
- A combined mode that outputs both the vital signs and position of a person; it can only be wall-mounted.

- The default detection distance settings are as follows:

x_min	-400 cm	x_max	400 cm
y_min	-100 cm	y_max	100 cm
z_min	40 cm	z_max	630 cm
alrmin	-75 °	alrmax	75 °
adumin	-75 °	adumax	75 °
Z always has + values			

- The smaller the object, the narrower the detectable angle. **The stable detection angle is -50° to 50°.**

- Vital-sign information can only be collected from the person closest to the sensor. If a moving object is closer to the sensor than the person whose vital signs are being detected, vital signs are not detected.
- Of the detected vital-sign information, the respiration rate measured in a stable state within 3 meters has an accuracy of 80% or more. Heart rate information has an accuracy of about 80%.



- Area A: heart rate, respiration, position
- Area B: respiration, position
- Area C: position

- **Detection may be impossible or accuracy may be reduced in the following cases:**

- Depending on the installation environment and the type of target, the actual detection distance may be shorter than the configured detection distance.
- The sensor detects vital signs by detecting the frequency of the small back-and-forth chest movements while a person breathes or the heart beats.
- Best performance is achieved in areas without objects such as furniture, for example meeting rooms or corridors. If there are complex objects in the detection area, detection may fail or inaccurate values may be output. People cannot be distinguished from moving objects, and people close to each other may be recognized as a single object. Running fans or air conditioners may also be recognized as objects.

- Data output format: ASCII

■ Mode without vital-sign waveform: default mode

&ABBBBCCCCDDDEEEFGGGHH! (23 byte) (default mode)		
Item	**Byte**	**Content**
&	1	Start character indicating combined vital-sign and distance data output
A	1	Number identifying the detected object. e.g. 1 - first object, 2 - second object
BBBB	4	X coordinate: left/right position relative to the radar, in cm. The first byte is the sign: '-' means left, '+' means right.
CCCC	4	Y coordinate: down/up position relative to the radar, in cm The first byte is the sign: '-' means down, '+' means up.
DDD	3	Z coordinate: front/back position relative to the radar, in cm.
EEE	3	Energy level (always 000 in the current version)
F	1	Vital-sign detection status 0: No target 1: Target present, moving 2: Target present, no vital signs 3: Target present, vital signs detected 4: Fall detected
GGG	3	Heart rate per minute
HH	2	Respiration rate per minute
!	1	End-of-data character
&0!		Output at the set time interval when no object is detected

- No sign is shown because only positive distances are output in the Z direction.

■ Mode with vital-sign waveform

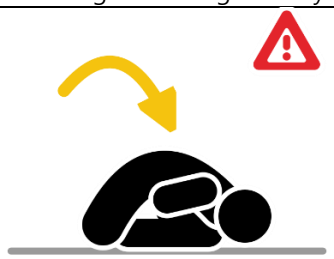
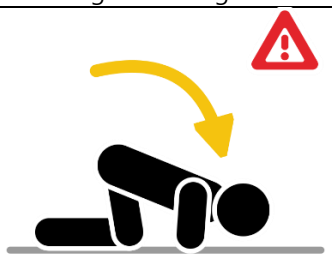
?ABBBBCCCCDDDEEEFGGGHHIJJJ! (27 byte)		
Item	**Byte**	
?	1	
A	1	Number identifying the detected object. e.g. 1 - first object, 2 - second object
BBBB	4	X coordinate: left/right position relative to the radar, in cm. The first byte is the sign: '-' means left, '+' means right.
CCCC	4	Y coordinate: down/up position relative to the radar, in cm The first byte is the sign: '-' means down, '+' means up.
DDD	3	Z coordinate: front/back position relative to the radar, in cm.
EEE	3	Energy level (always 000 in the current version)
F	1	Vital-sign detection status 0: No target 1: Target present, moving 2: Target present, no vital signs 3: Target present, vital signs detected 4: Fall detected
GGG	3	Heart rate per minute

HH	2	Respiration rate per minute
I	1	Heartbeat pulse, 0, 1
JJJ	3	Respiration waveform data, 0 ~ 999 (3 digits; leading digits padded with spaces instead of 0)
!	1	
?0!		Output at the set time interval when no object is detected

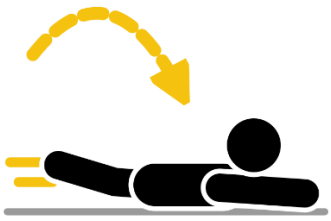
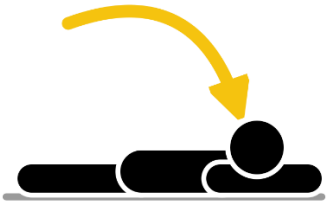
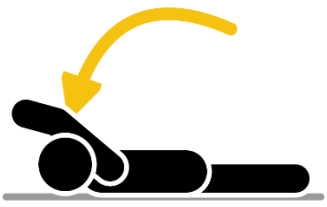
- The initial state of this mode is the basic data mode without vital-sign waveform data. To include vital-sign waveform data, use 'wave 1₩n' for additional settings.

- **Fall alerts do not cover every situation; they respond only to common everyday falls such as the following.**
  (Performance is currently being updated; this function is provided at demo level.)

■ Common fall situations to which this sensor responds

Collapsing in place	Crouching and falling sideways	Crouching and falling forward
		

■ **Uncommon situations to which this sensor does not respond**

Sliding on the floor while jumping	Falling forward with the body straight	Falling backward with the body straight
		

■ Normal situations to which this sensor does not respond

Sitting on a chair	Sitting on the floor	Lying on the floor
		

For the fall alert function to work properly, the sensor height above the floor must be set correctly. In mode 6, the command for setting the sensor height is fallsh, and its default value is 90.

7.7 Option mode: Advanced Vital Sign mode (mode 11)

- Can be enabled and used with an additional license.
- The intelligent vital-sign detection mode is an advanced mode based on the basic vital-sign detection mode (mode 5). It includes RMSSD and SDNN data, which are components of HRV
- RMSSD evaluates parasympathetic nervous activity (cardiac stability and recovery) and measures short-term heart-rate variability. The result is output when vital signs have been detected continuously in a stable state for 30 seconds or more.
- SDNN reflects the balance between the sympathetic and parasympathetic nervous systems; data output starts when vital signs have been detected in a stable state for 30 seconds or more. SDNN is based on short-term monitoring and stabilizes to a normal value after a stable state has been maintained for at least 5 minutes.

- The default detection range settings are as follows (similar to mode 5, but the y-axis range differs):

x_min	-125 cm	x_max	125 cm
y_min	-100 cm	y_max	100 cm
z_min	40 cm	z_max	250 cm
alrmin	-75 °	alrmax	75 °
adumin	-75 °	adumax	75 °
Z always has + values			

The stable detection angle is -50° to 50°.

- This mode supports the extended data mode. The extended data mode is enabled with the command 'exten 1₩n' and disabled with 'exten 0₩n'.
- When the extended mode is enabled, the following additional data are provided:

Normal mode	Vital status Heart rate Breath rate RMSSD data SDNN data Heart wave data Breath wave data Option data : distance
Extension mode	This mode includes all data of the normal mode and provides additional information. X-axis distance information Y-axis distance information Z-axis distance information Reserved data (always 0)

- Data output format: ASCII
- Output mode with waveform: default mode: output 10 times per second

Basic data mode		
{ABBBCCDDEEFGGGHHH! (19 byte)		
Item	**Byte**	**Content**
{	1	Start character indicating vital-sign data output
A	1	Vital-sign detection status 0: No target 1: Target present, moving 2: Target present, no vital signs 3: Target present, vital signs detected
BBB	3	Heart rate per minute
CC	2	Respiration rate per minute
DD	2	HRV: RMSSD (ms), 0 ~ 99
EE	2	HRV: SDNN (ms), 0 ~ 99
F	1	Heartbeat pulse, 0, 1

GGG	3	Respiration waveform data, 0 ~ 999
HHH	3	Detection distance (cm)
!	1	End-of-data character
{0!		Output at the set time interval when no object is detected
Extended data mode		
{ABBBCCDDEEFGGGHHHHIIIIJJJKKK! (30 byte)		
Item	**Byte**	**Content**
{	1	Start character indicating vital-sign data output
A	1	Vital-sign detection status 0: No target 1: Target present, moving 2: Target present, no vital signs 3: Target present, vital signs detected
BBB	3	Heart rate per minute
CC	2	Respiration rate per minute
DD	2	HRV: RMSSD (ms), 0 ~ 99
EE	2	HRV: SDNN (ms), 0 ~ 99
F	1	Heartbeat pulse, 0, 1
GGG	3	Respiration waveform data, 0 ~ 999
HHHH	4	X coordinate: left/right position relative to the radar, in cm. The first byte is the sign: '-' means left, '+' means right.
IIII	4	Y coordinate: down/up position relative to the radar, in cm. The first byte is the sign: '-' means down, '+' means up.
JJJ	3	Z coordinate: front position relative to the radar, in cm.
KKK	3	Reserved data (always 0)
!	1	End-of-data character
{0!		Output at the set time interval when no object is detected

■ Output mode without waveform

Basic data mode		
}ABBBCCDDEEFFF! (15 byte) (default mode)		
Item	**Byte**	**Content**
}	1	Start character indicating vital-sign data output
A	1	Vital-sign detection status 0: No target 1: Target present, moving 2: Target present, no vital signs 3: Target present, vital signs detected
BBB	3	Heart rate per minute
CC	2	Respiration rate per minute
DD	2	HRV: RMSSD (ms), 0 ~ 99
EE	2	HRV: SDNN (ms), 0 ~ 99
FFF	3	Detection distance (cm)
!	1	End-of-data character
}0!		Output at the set time interval when no object is detected
Extended data mode		
}ABBBCCDDEEFFFFGGGGHHHHII! (26 byte)		
Item	**Byte**	**Content**
}	1	Start character indicating vital-sign data output
A	1	Vital-sign detection status 0: No target 1: Target present, moving 2: Target present, no vital signs 3: Target present, vital signs detected

BBB	3	Heart rate per minute
CC	2	Respiration rate per minute
DD	2	HRV: RMSSD (ms), 0 ~ 99
EE	2	HRV: SDNN (ms), 0 ~ 99
FFFF	4	X coordinate: left/right position relative to the radar, in cm. The first byte is the sign: '-' means left, '+' means right.
GGGG	4	Y coordinate: down/up position relative to the radar, in cm. The first byte is the sign: '-' means down, '+' means up.
HHH	3	Z coordinate: front position relative to the radar, in cm.
III	3	Reserved data (always 0)
!	1	End-of-data character
}0!		Output at the set time interval when no object is detected

■ The initial state of this mode is the basic data mode without vital-sign waveform data. To include vital-sign waveform data, use 'wave 1₩n', and to output extended data, use 'exten 1₩n' for additional settings.

7.8 Restroom ceiling mounting mode (mode 12)

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- A dedicated mode installed on the ceiling of a restroom stall that detects a person's presence, posture (standing / sitting on the toilet / lying down), respiration rate per minute and continuous stay time. The default detection area is set narrower than in other ceiling modes to suit the small space.

- The default detection distance settings are as follows:

x_min	-50 cm	x_max	50 cm
y_min	-250 cm (based on rrsh 250cm)	y_max	-20 cm
z_min	-40 cm	z_max	100 cm
alrmin	-25 °	alrmax	25 °
adumin	-25 °	adumax	25 °
Sensor height (rrsh)	Factory default 250 cm (reconfigurable within 100~300cm)		
Y is always negative (-).			

- The sensor installation height must be reset to the actual installation height (100~300cm) with the rrsh command. The factory default is 250cm; entering the rrsh command in other modes returns =UnSupport!.
- If detection is lost for about 0.8 s or more after an approach near the entrance has been confirmed, the state switches to vacant immediately. Other losses (e.g. a momentary loss of detection while the person stays still) keep the last state for 10 minutes before switching to vacant. If the actual entrance position is not set accurately with rrdoorx/rrdoorz/rrdoorwd, all exit decisions may be delayed.

- **Detection may be impossible or accuracy may be reduced in the following cases:**
- Because of the narrow stall, movements of the door or partition, ventilation fans and the like may be mistaken for occupancy. This can be solved by registering those locations as detection exclusion zones with exzon/exzset.
- If two or more people are in the stall at the same time, the status is output only as 'two or more people detected', and the posture of each person is not distinguished.
- The exten, hmax and bmax commands do not affect the output of this mode (hmax/bmax are rejected with =UnSupport! when set). The wave command is an exception: if wave 1 is set while the posture is sitting/lying and not in a warning state, the output switches to continuous output.

- Data output format: ASCII

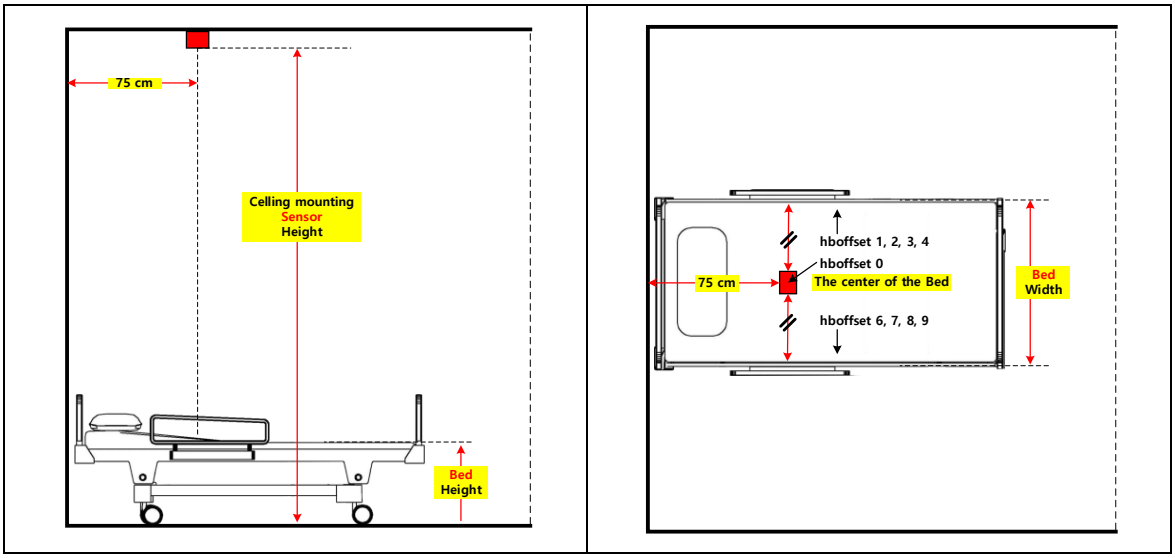
Restroom status output mode		
^1SXZTTBBB0000! (15 bytes) (when occupied)		
Item	**Byte**	**Content**
^	1	Data start character of the restroom status output mode
1	1	Presence flag (always '1'). When vacant, the ^0! short packet below is output instead.
S	1	Restroom status / 1: standing / 2: sitting on the toilet / 3: lying down (suspected collapse) / 4: two or more people detected / 5: warning (abnormal situation such as proximity to the ceiling) / 6: stay time exceeded (rstc setting elapsed)
X	1	X zone (1~3: left/center/right relative to the radar)
Z	1	Z zone (1~3: near/middle/far relative to the radar)
TT	2	Continuous stay time (minutes). Fixed at the maximum of 99.
BBB	3	Respiration rate per minute. Fixed at the maximum of 99.
0000	4	Reserved field (always 0).
!	1	End-of-data character.
^0!	^0!	Short packet output when the restroom is empty (vacant) (3 bytes).

7.9 Option mode: Hospital bed mode – Ceiling mounting (mode 13)

- Can be enabled and used with an additional license.

- Hospital bed mode is an integrated mode that monitors the patient's presence, posture, vital signs, fall risk and pressure-ulcer risk together. The default output starts with '/'; when wave 1 is enabled, output starts with '₩' and includes the respiration waveform.
- The installation is fixed to ceiling mounting. The basic procedure is to set the sensor installation height (200~300 cm) with hbsh, then combine hbh, hbw, hboffset, hbwl and hbwr to align the bed center and wall direction. hbdl and hbdr are filters that reject reflections higher than the bed, such as curtains.
- In ceiling mounting mode, when the bed touches the wall with the head toward the wall, install the sensor about 75 cm (fixed value) from the wall. Align the sensor with the exact center of the bed in the left-right direction.
- The main configuration commands are hbsh, hbh, hbw, hboffset, hbwl, hbwr, hbdl, hbdr, hbdlh, hbdll, hbdrh, hbdrl, hbfalla, hbfalls, hbedsorea, hbedsores, hrv, wave and interval. The hbsh range is 200~300 cm, the hbh range is 10~100 cm (but no more than hbsh - 100 cm), and the hbw range is 80~170 cm. The head/foot detection range itself is always fixed to the inside of the bed and there is no command to turn it off entirely; only the four corners combined with left/right (for excluding fixed furniture such as wardrobes) can be excluded individually with hbdlh/hbdll/hbdrh/hbdrl.
- The recommended order is: enter mode 13 -> set the sensor height with hbsh -> set hbh, hbw, hboffset, hbwl/hbwr -> set hbdl/hbdr (curtain height filter) if needed -> set corner (wardrobe) exclusions with hbdlh/hbdll/hbdrh/hbdrl if needed -> set display options and wave if needed -> check the status packets.

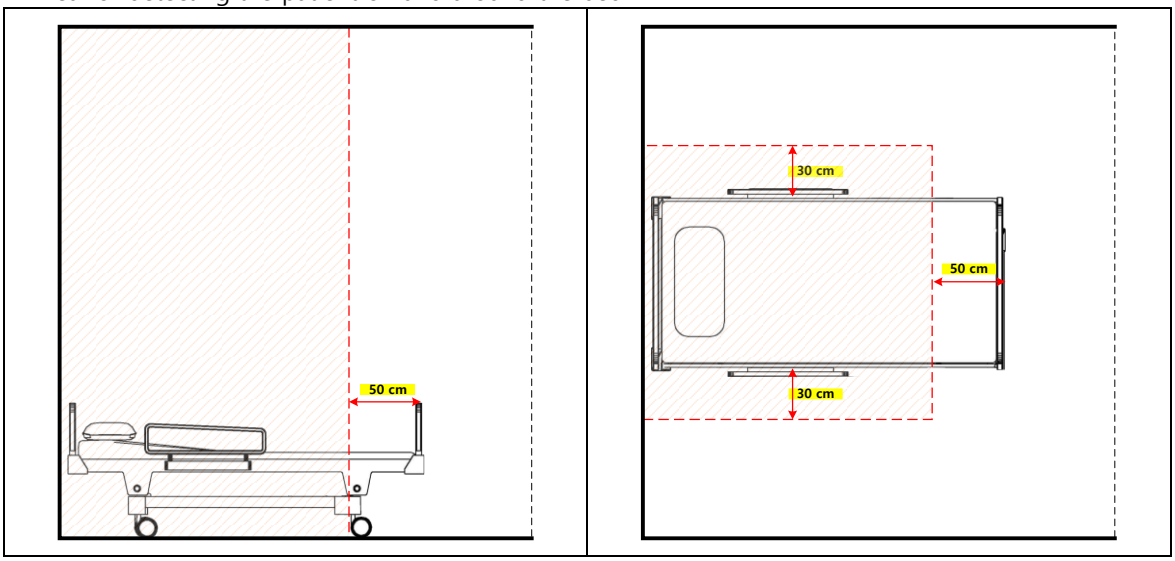
- The installation position of the AS_613 radar sensor in ceiling mode is as follows:



- In ceiling mounting mode, when the bed touches the wall with the head toward the wall, the sensor must be installed 75cm from the wall.

- The sensor must be exactly at the center of the bed in the left-right direction. If the bed moves left or right within 20cm after the sensor is installed, adjust the offset with the 'hboffset' command.

■ Area for detecting the patient on and around the bed



- The detection area includes the entire area inside the bed and extends 45cm beyond the bed to prevent other patients or medical staff moving outside the bed from being detected.

■ The fall-risk warning is activated when the patient, lying or sitting in the center of the bed, sits up to the left or right side, indicating an intention to get up and leave the bed.

■ The fall alarm is activated when, in the fall-risk state, the patient is detected sitting or lying on the floor beside the bed.

■ A fall is detected only when the patient, lying in bed, sits on the edge of the bed and then sits or lies on the floor. Falls are not detected in other situations.

- **When the patient is lying in bed, respiration, heart rate and, depending on the option, waveforms are output. While lying in bed, the patient may take various postures, not only lying straight but also with hands on the chest or lying on the side. Because the radar does not determine the posture of the patient in bed, it cannot tell whether the detected heart rate was measured in a normal state. Therefore, the heart rate can be measured but should be used for reference only. Respiration and waveforms, on the other hand, are more accurate than heart rate. In the current version, only the respiration waveform is provided**.

- The default detection distance settings are as follows:

x_min	-90 cm	x_max	90 cm
y_min	-220 cm	y_max	-50 cm
z_min	-65 cm	z_max	130 cm
alrmin	-55 °	alrmax	55 °
adumin	-45 °	adumax	55 °
Sensor height	220 cm	Bed height	55 cm
		Bed width	90 cm
Y always has - values			

- The bed width (hbw) setting range is 80 cm ~ 170 cm (default 90 cm).

- Data output format: ASCII
- Mode without vital-sign waveform: default mode

/A! (when the state of A is 0, 1, 2 or 7)
/ABBBCCDDEEFFGGHIIJJJ! (22 byte) (default mode)

Item	Byte	Content
/	1	Start character indicating data output without vital-sign waveform
A	1	Hospital bed presence status 0: Bed empty 1: Patient standing beside the bed 2: Patient sitting or lying beside the bed 3: Patient in bed (awake) 4: Patient in bed (tossing and turning) 5: Patient in bed (asleep) 6: **Patient in bed (fall-risk warning)** 7: Fall detected 8: Pressure-ulcer risk warning 9: High pressure-ulcer risk warning
BBB	3	Heart rate per minute
CC	2	Respiration rate per minute
DD	2	SpO2; value output only when additional equipment is connected 00: No additional equipment connected
EE	2	Body temperature; value output only when additional equipment is connected 00: No additional equipment connected
FF	2	HRV: RMSSD (ms), 0 ~ 99
GG	2	HRV: SDNN (ms), 0 ~ 99
H	1	Type of additionally connected equipment 0: No additional equipment connected
II	2	Status of the connected equipment 00: No additional equipment connected
JJJ	3	Reserved data (always +00)
!	1	End-of-data character
/0!		Output at the set time interval when no object is detected

■ Mode with vital-sign waveform: output 10 times per second

₩A! (when the state of A is 0, 1, 2 or 7) **₩ABBBCCDDEEFFGGHIIJJJKKK!** (25 byte)		
Item	**Byte**	**Content**
₩	1	Start character indicating vital-sign waveform data output
A	1	Hospital bed presence status 0: Bed empty 1: Patient standing beside the bed 2: Patient sitting or lying beside the bed 3: Patient in bed (awake) 4: Patient in bed (tossing and turning) 5: Patient in bed (asleep) 6: **Patient in bed (fall-risk warning)** 7: Fall detected 8: Pressure-ulcer risk warning 9: High pressure-ulcer risk warning
BBB	3	Heart rate per minute
CC	2	Respiration rate per minute
DD	2	SpO2; value output only when additional equipment is connected 00: No additional equipment connected
EE	2	Body temperature; value output only when additional equipment is connected 00: No additional equipment connected
FF	2	HRV: RMSSD (ms), 0 ~ 99
GG	2	HRV: SDNN (ms), 0 ~ 99
H	1	Type of additionally connected equipment 0: No additional equipment connected
II	2	Status of the connected equipment 00: No additional equipment connected
JJJ	3	Reserved data (always +00)
KKK	3	Respiration waveform data
!	1	End-of-data character
₩0!		Output at the set time interval when no object is detected

7.10 Output data start and end characters for each mode

Order	Mode	Data output start character
1	Mode 1: Wall mounting position measurement mode	# : Distance output mode
2	Mode 2: Ceiling mounting position measurement mode	$: Distance output mode
3	Mode 3: Wall mounting presence (occupancy) detection mode	* : Presence/occupancy-grid output mode
4	Mode 4: Ceiling mounting presence (occupancy) detection mode	~ : Presence/occupancy-grid output mode
5	Mode 5: Wall mounting vital-signal mode	% : Without vital waveform / @ : Including vital waveform
6	Mode 6: Wall mounting vital+position mode	& : Without vital waveform / ? : Including vital waveform
7	Mode 11: Advanced vital sign mode	} : Without vital waveform / { : Including vital waveform
8	Mode 12: Restroom ceiling mode	^ : Restroom state output mode
9	Mode 13: Hospital bed ceiling mode	/ : Without vital waveform / ₩ : Including vital waveform
10	Response to Command	=
11	End character of all packets	!

8 Protocol

- AS_613 supports 3.3V-level UART communication and can interface with 1.8V, 3.3V and 5V signals.
- ASCII character codes are used for data transmission and reception, and Command Line Input (CLI) is supported.
- The LED turns on when an object is detected.
- Whether the LED lights up can be set with the led command (default: enabled); identification blinking with the wink command always works regardless of this setting.

- UART settings:

Item	Setting value
Baud rate	460,800 bps (default); 9,600 / 19,200 / 38,400 / 57,600 / 115,200 / 230,400 / 460,800 / 921,600 supported
Data	8 bit
Parity bit	None
Stop bits	1 bit
Flow control	None

- All commands must be entered in lowercase.

- Settings stored in NV keep their values after power cycling. Runtime commands such as start, stop and pulse are not retained.
- Every command must be sent with LF (line feed/new line) or CR (carriage return).
- The start character of a command response is '=' and the end character is '!'.
  e.g. the response to the standby command is =15!.
- **Some configuration commands automatically perform a reset or fast reset internally, while others take effect immediately. Follow each command's description for how changes are applied.**
- When sending consecutive commands, it is recommended to confirm the response to the previous command before sending the next one. After commands that trigger an automatic reset or fast reset, allow time for the system to restart.

- **In host implementations, it is safe to allow a margin of several hundred ms or more after commands that trigger an automatic reset.**
- **It is recommended to decide the installation orientation commands (mode, rotate, walltilt) first, and then tune the xmin~zmax detection area and exz* (exclusion zone) commands. If the orientation commands are changed later, coordinates set earlier may be misaligned in the new coordinate system.**

- **USB settings:**
- When connected to a PC through the USB-C connector (J1), AS_613 operates as a USB CDC device (virtual COM port) and uses the same commands and response formats as UART.

- When connected to a PC, it is recognized automatically by the built-in Windows driver without installing a separate driver, and appears in Device Manager as 'USB Serial Device (COMxx)'.

- The baud rate set with the baud command does not apply to the USB connection, which runs at USB speed. The baud setting applies only to UART (TX/RX pins) communication.

- UART and USB receive commands independently, and all responses and detection data are output to both interfaces simultaneously.

- When several modules are connected to one PC at the same time, each module is identified by its USB serial number (AS613-000 ~ AS613-009), and up to 10 modules can be recognized. The default is assigned automatically from the module's unique ID; if the automatic values of two modules overlap, the number of one can be assigned manually with the usbid command.

- The USB device identifiers are VID (Vendor ID) 0x04B4 and PID (Product ID) 0x0613.

9 Sensing Exclusion Zone Setting Features

- The detection exclusion zone function (Exclusion Zone) can store up to 5 exclusion zones on the XZ plane for each combination of active mode and installation/operation, and is configured with the exzon, exzinfo, exzset, exzpick and exzdel commands.

- The detection exclusion zone function specifies rectangular exclusion zones on the radar's XZ plane to exclude fixed reflectors such as fans, air conditioners and furniture from the output and presence decisions.
- Up to 5 exclusion zones can be specified per mode, and different installation positions or operating modes are managed independently. Therefore, set the mode and installation/operation combination you will use first, and then configure the exclusion zones.

- There are two ways to set detection exclusion zones:
- the exzpick command, which sets an exclusion zone automatically based on the position of a currently detected target,
- and the exzset command, with which the user specifies the position and range directly.

- After setting, the exzinfo command shows the number of stored slots, or the position (cx, cz) and exclusion range (hx, hz) of each slot.
- The function can also be enabled and disabled. The exzon 1 command enables the exclusion zone function for the currently active profile. Entering exzon alone queries the current state, and exzon 0 disables only the function while keeping the stored slot data.
- To delete a specific slot, use exzdel <slot number>; to delete the whole active profile, use exzdel 0. exzdel 0 deletes all slots of the currently active profile and also turns the function OFF.

- The procedure can be summarized as: complete the desired mode settings → register exclusion zones with exzset or exzpick → check the stored contents with exzinfo → enable with exzon 1.

- cx : X-coordinate center of the exclusion zone, in cm
- cz : Z-coordinate center of the exclusion zone, in cm
- hx : half-width of the exclusion zone in the X direction, in cm,
  e.g. 40 sets the area from -40cm to the left to 40cm to the right of the cx coordinate as the exclusion zone
- hz : half-width of the exclusion zone in the Z direction, in cm,
  e.g. 50 sets the area from -50cm to 50cm around the cz coordinate as the exclusion zone

Command	Format	Main responses	Description and notes
exzon	exzon exzon 1 exzon 0	=0!, =1! =OverRange!	Queries or turns ON/OFF the exclusion zone function in the current mode. 0: Keep zone settings and disable only the function 1: Enable the detection exclusion zone function
exzinfo	exzinfo exzinfo <slot>	=N! =cx,cz,hx,hz! =OverRange!	Queries the number of stored active slots (N=0~5) or the coordinates/half-widths of a specified slot. slotis 1~5; 0,0,0,0 may be shown when querying an empty slot.
exzset	exzset <cx> <cz> exzset <cx> <cz> <hx> exzset <cx> <cz> <hx> <hz>	=cx,cz,hx,hz! =OverRange!	Registers coordinates directly in an empty slot of the currently active profile. <hx> <hz>If omitted, <hx> <hz> default to 40; if only <hx> is entered, the <hx> value is also applied to <hz> After registration, slots are automatically

			sorted by cz in ascending order Outputs =OverRange! if there is no empty slot or the parameter format/value is invalid.
exzpick	exzpick <target_num> exzpick <target_num> <hx> exzpick <target_num> <hx> <hz>	=target_num,cx,cz,hx, hz! =Missing! =NoTarget! =OverRange!	Reads the position of the currently output target number (1~5) and registers it automatically in an empty slot. <hx> <hz>If omitted, <hx> <hz> default to 40; if only <hx> is entered, the <hx> value is also applied to <hz> =Missing! if the target number does not exist, =NoTarget! if the specified target does not exist, =OverRange! if there is no empty slot or the format is invalid
exzdel	exzdel <slot> exzdel 0	=slot! =0! =OverRange!	slot 1~5 deletes an individual slot; 0 deletes all slots of the currently active profile and then disables the function Slots are automatically re-sorted after deletion

9.1 Operation summary by mode (based on current firmware)

■ The table below summarizes, based on the current firmware, the purpose, main commands, recommended order of use and precautions for each mode for quick reference.

Mode	Main commands	Recommended order of use	Precautions
Mode 1 DISTANCE_WALL Wall mounting distance/coordinate measurement Default #, vacant #0!	mode 1 exten x/y/z alr/adu wait sena/senb walltilt senin/senout rotate exz*	Enter mode 1 Adjust ROI/angles Adjust sensitivity Check packets	Wall mounting only First output may be affected by the previous buffer
Mode 2 DISTANCE_CEILING Ceiling mounting distance/coordinate measurement Default $, vacant $0!	mode 2 exten x/y/z alr/adu wait sena/senb senin/senout/gnd rotate exz*	Enter mode 2 Adjust ceiling ROI Adjust sensitivity Check packets	Ceiling mounting only Position measurement mode, not a vital-sign mode
Mode 3 PRESENCE_WALL Wall mounting presence (occupancy) detection Detects 8m x 8m in 256 zones Default *, vacant *0!	mode 3 x/y/z alr/adu sena/senb walltilt senout (20~500 s, 10 s steps) exzon/exzset	Enter mode 3 Adjust ROI Adjust sensitivity Set zone hold time with senout Check packets	Zone (block)-based detection without the coordinate tracker senin/hmax/bmax/wave/exten/interval do not affect the output of this mode exzpick returns =UnSupport! (use exzset)
Mode 4 PRESENCE_CEILING Ceiling mounting presence (occupancy) detection Detects 8m x 8m in 256 zones Default ~, vacant ~0!	mode 4 x/y/z alr/adu sena/senb/op senout (20~500 s, 10 s steps) exzon/exzset	Enter mode 4 Adjust ceiling ROI Adjust sensitivity Set zone hold time with senout Check packets	Zone (block)-based detection without the coordinate tracker gnd can be set/queried but does not affect the decision senin/hmax/bmax/wave/exten/interval do not affect the output of this mode exzpick returns =UnSupport!
Mode 5 VITAL Wall mounting vital sign	mode 5 wave interval hmax/bmax	Enter mode 5 Adjust ROI/hmax/bmax wave 1 if needed exten 1 if needed	Strongly affected by the nearest movement Apnea is not a separate status code; it appears as a respiration rate close to 00 interval cannot be changed during wave 1

Default %, wave=1 -> @	exten geometry sena/senb/op senin/senout exz*		
Mode 6 VITAL_POSITIO N Wall mounting combined vital sign + position Default &, wave=1 -> ?	mode 6 wave interval hmax/bmax geometry sena/senb walltilt senin/senout exz*	Enter mode 6 Adjust wall ROI Check vital sign/position based on the first slot	Vital-sign data is meaningful only in the first slot Fall AI state may be exposed in the status field
Mode 11 ADV_VITAL Advanced vital sign Default }, wave=1 -> {	mode 11 wave interval hmax/bmax exten geometry sena/senb/op senin/senout exz*	Enter mode 11 Adjust vital-sign ROI Measure for a long time in a stable posture	RMSSD/SDNN require a stable period wave/interval relationship is the same as mode 5 Default geometry differs from mode 5 (y -100~100cm)
Mode 12 RESTROOM Ceiling mounting restroom status detection Output ^, vacant ^0!	mode 12 rstc rrsh rrdoorx/rrdoorz/ rrdoorwd geometry sena/senb/op/g nd senin/senout exz*	Enter mode 12 Set sensor height with rrsh Adjust ceiling ROI Set rstc Set the entrance with rrdoorx/rrdoorz/rrdoor wd Check status packets	exten/wave/hmax/bmax have no effect rrsh factory default is 250cm and must be reset to the actual installation height Exit decisions may be delayed if the entrance is not set to measured values
Mode 13 HOSPITAL_BED Bed presence/postu re/vital signs/fall/press ure ulcer Default /, wave=1 -> ₩	mode 13 hbsh hbh/hbw/hboffs et hbwl/hbwr hbdl/hbdr hbdlh/hbdll/hbd rh/hbdrl hbfalla/hbfalls hbedsorea/hbed sores hrv sena/senb/op/g nd wave interval	Enter mode 13 Set sensor height with hbsh Adjust bed parameters Set display options Check status packets	Fixed to ceiling mounting (wall mounting not supported, rotate fixed at 180°) hbsh 200~300, hbh 10~100 (but no more than hbsh-100), hbw 80~170 Head/foot directions are always detected only up to the inside of the bed (no disable command; only corners can be excluded individually with hbdlh, etc.)

Response type	**Meaning**
=<value>!	Query or setting completed normally
=OverRange!	Error in the number, range or format of arguments
=UnSupport!	Command not supported in the current mode/mount/work combination
=UnKnown!	Unregistered command, or failure to register the request in the internal

	queue
=Busy!	The internal command queue is full and cannot accept the request — confirm the response to the previous command and send again.
=Error!	Internal processing failure, such as a Flash write

10 Command

Command	Example	Default value / Setting range	Explanation
Flow / power control commands			
start	start	-	Starts data output. If start is sent right after stop, the first packet may contain previous values. When executed in pwr 0 (standby mode), the module returns to the power level used before entering standby (=Start!) and restarts internally, taking about 1 s to take effect.
stop	stop	-	Stops data output only. Detection and state accumulation continue internally.
pwr	Pwr pwr 2	2 / 0 ~ 2	Sets or queries the power mode (response =Pwr 0!/1!/2!). 0: Standby (sleep) mode — completely cuts the MMIC (radar sensor) LDO power and stops all detection. The LED, USER_IO, GPIO1 and GPIO2 turn off immediately regardless of the uiosec/ledsec hold-time settings. 1: Power-saving mode — about half of the maximum output; maximum detection distance about 86% of pwr 2. 2: Maximum performance mode (default). Standby can be exited only with start or pwr 1/2; the module restarts internally and takes about 1 s to take effect. Levels 1/2 are stored in NV and kept after reboot, but 0 (standby) is not stored, so if power is cycled during standby the module starts at the last active level (1 or 2).
reset	reset	-	Restarts the module in software while keeping the settings.
freset	freset	-	Factory reset. Restores all NV settings — operating mode, detection area (x/y/z ranges, detection angles), sensitivity (sena/senb/senin/senout/op/gnd), output interval (interval), wait time (wait), extended output (exten), vital-sign distance limits (hmax/bmax), exclusion zones (exz), power mode (pwr), etc. — to factory defaults and then restarts. If you have tested with many changed settings, previous settings may remain and cause unintended behavior, so freset is recommended before starting a new configuration. To restore only the settings of the current mode, use mreset.
mreset	mreset	-	Restores only the settings of the currently active mode to factory defaults. Settings of other modes are kept. Takes effect immediately without a reboot (=mReset!).
cfgdump	cfgdump	-	Outputs a backup of all currently stored settings as 32 lines (each line =CFG:NN:16-byte hex!). Device-specific identifiers such as run_count/fault_count/app_baud/radar_fault_cnt/usb_id are masked as 0, so the values can safely be transferred as-is to another device.
cfgload	cfgload	-	Receives, in order, the 32 lines backed up with cfgdump and restores all settings. Each line must be in the format =CFG:NN:16-byte hex!, and the operation is cancelled if a line is not entered within 10 s (=CfgLoadTimeout!). When all 32 lines are received correctly, it outputs =CfgLoadDone:mode:NN! and reboots.
exzdump	exzdump	-	Outputs a backup of only the currently stored detection exclusion zone settings as 32 lines (each line =EXZ:NN:16-byte hex!).
exzload	exzload	-	Receives, in order, the 32 lines backed up with exzdump and restores only the detection exclusion zone settings. The format and timeout rules are the same as cfgload, but unlike

			cfgload it takes effect immediately without a reboot (=ExzLoadDone!).
Diagnostics / firmware management commands			
ver	ver	-	Shows the firmware version and build date. e.g. =L61326Oct26.13.V.3.0!
info	Info info 10	- / 1 ~ 100 (s)	Without parameters, outputs all current settings in blocks such as =INFO:, =GEO:, =SEN:, =OUT:. With a value of 1~100 seconds, streams real-time coordinates for that time.
wink	wink	-	Blinks the LED for about 1.5 s (10 times at 150ms intervals) for product identification. Always operates at maximum brightness regardless of the led setting.
led	led led 5	5 / 0 ~ 9	0: LED disabled (always off). 1~9: LED enabled (1 is dimmest and 9 brightest; steps are calibrated to perceived brightness). Blinking by wink always operates at maximum brightness, even with led 0.
ledsec	Ledsec ledsec 5	0 / 0 ~ 999 (s)	LED output hold time. 0: on/off immediately with target detection/loss (default). 1~999: stays on for the set number of seconds after the target disappears, then turns off automatically. This value is not stored in NV and is always reset to 0 on reboot.
uiosec	Uiosec uiosec 5	0 / 0 ~ 999 (s)	Output hold time of the USER_IO pin. 0: on/off immediately with target detection/loss (default). 1~999: keeps the output for the set number of seconds after the target disappears, then turns off automatically. When entering pwr 0 (standby mode), it turns off immediately regardless of the hold-time setting.
check	check	-	Checks the module's power and operating status. It checks, in order, the input power (5V, under load) → radar IC (MMIC) response → TX output, and responds =CHECK:OK! if all are normal, or =CHECK:FAIL:item,item! listing only the failed items. Normal detection stops for the few seconds of the check and is automatically re-initialized afterwards. It is rejected with =Pwr0Active! in standby mode (pwr 0) and with =RadarHalted! when the radar has stopped abnormally.
update	update	-	Stops data streaming and enters the bootloader for firmware update (the update procedure proceeds after the =OK! response).
Mode / communication setting commands			
mode	mode₩ mode 5	1 1, 2, 3, 4, 5, 6, 11, 12, 13	Sets or queries the sensor's operating mode. 1: wall mounting position measurement, 2: ceiling mounting position measurement, 3: wall mounting presence detection, 4: ceiling mounting presence detection, 5: wall (side) mounting vital sign, 6: wall mounting combined vital sign + position, 11: advanced vital sign (option), 12: restroom ceiling mounting (option), 13: hospital bed ceiling mounting (option). After the value is changed, the sensor switches to the new mode a few frames later; values not defined for the product's shipping configuration return =UnDefine!.
baud	Baud baud 460800	460800 / 9600, 19200, 38400, 57600, 115200, 230400, 460800, 921600	Sets the UART (TX/RX pins) baud rate. The module restarts immediately when the value is changed. This setting does not apply to the USB connection; USB runs at its own speed.
usbid	Usbid usbid 7	Automatic (based on module unique ID) / 0 ~ 9	Queries or manually sets the last digit of the USB serial number (AS613-00<N>). Used to distinguish modules with different values when several are connected to one PC at the

			same time. When a value is set, it is stored in NV and the device restarts automatically; the COM port number may change on reconnection.
Output / vital-sign setting commands			
exten	exten exten 1	0 / 0, 1	Extended output ON/OFF. Applies only to the output packets of modes 1, 2, 5 and 11. In modes 3/4 (presence detection), mode 6 (vital sign + position), mode 12 (restroom) and mode 13 (hospital bed), the value is stored but does not affect the output (there are no fields it applies to).
interval	interval interval 1	1 / 1 ~ 255	Sets the data output time interval. The maximum data output rate is 10 times per second. Data is output once every set number of frames. 1: output every frame, 10 times per second 2: output every second frame, 5 times per second 10: output every tenth frame, once per second In vital-sign output modes, setting wave 1 fixes the output to real time (1). In modes 3/4 (presence detection), the value is stored but does not affect the output (a separate scheduler determines when to send).
pulse	pulse 1 pulse 0	0 / 0, 1	Turns on/off the pulse signal output synchronized with respiration detection. This is a runtime setting and is not stored.
wait	wait wait 15	15 / 3 ~ 43200 (s), 0 = restore default	Time interval at which the vacant (0) packet is re-output when no target is detected.
wave	wave wave 1	0 / 0, 1	Turns waveform output on/off in vital-sign output modes (5, 6, 11, 12, 13). 1: waveform output ON and output fixed to real time (10 times per second). 0: waveform output OFF; returns to the stored interval. Trying to set the value in other modes (1, 2, 3, 4) returns =UnSupport!. However, a query without parameters (wave) outputs the last stored value regardless of the current mode.
hmax	hmax hmax 250	250 / 40 ~ 600cm, 5cm steps) && (<= bmax)	Sets the upper distance limit for heartbeat detection. Only for modes 5, 6 and 11; trying to set the value in other modes (including restroom and hospital bed modes) returns =UnSupport!. However, a query without parameters (hmax) outputs the last stored value regardless of the current mode.
bmax	bmax bmax 300	300 / (40 ~ 600cm, 5cm steps) && (<= zmax)	Sets the upper distance limit for respiration detection. Only for modes 5, 6 and 11; trying to set the value in other modes returns =UnSupport!. However, a query without parameters (bmax) outputs the last stored value regardless of the current mode.
Detection area setting commands			
xmin / xmax	xmin -200 / xmax 250	Wall -450 ~ 450cm / Ceiling -450 ~ 450cm	Left-right (X-axis) detection boundary. In hospital bed mode (13) it is determined automatically from the bed width (hbw), so it returns =UnSupport!.
ymin / ymax	ymin -125 / ymax 100	Wall -450 ~ 450cm / Ceiling: ymin -450~0cm, ymax -300~0cm	Up-down (Y-axis) detection boundary. Ceiling mounting always allows only negative (-) Y values. In restroom mode (12) and hospital bed mode (13) it is determined automatically from the rrsh/hbsh sensor installation height, so it returns =UnSupport!.
zmin / zmax	zmin 40 / zmax 600	Wall 0 ~ 900cm (vital-sign modes 0~630cm) / Ceiling -450 ~ 450cm	Front-back (Z-axis, distance) detection boundary. Wall mounting modes always allow only positive (+) Z values.
alrmin /	alrmin -75	-90 ~ 0 / 0 ~ 90 (°)	Left-right (horizontal) detection angle. Left of the radar is -°,

alrmax	/ alrmax 75		right is +°.
adumin / adumax	adumin -75 / adumax 75	-90 ~ 0 / 0 ~ 90 (°)	Up-down (vertical) detection angle. Below the radar is -°, above is +°.
walltilt	walltilt walltilt 30	0 / 0 ~ 45 (°)	In wall mounting modes (1, 5, 6, 11), compensates for the angle when the sensor is tilted downward rather than installed horizontally (0°). The x/y/z coordinates are corrected by the set angle. In other modes/installations, including mode 3 (wall mounting presence detection), it returns =UnSupport!.
rotate	rotate rotate 90	0 / 0, 90, 180, 270	Transforms coordinates when the sensor is installed rotated clockwise. In hospital bed mode (13), a 180° rotation is fixed internally, so it returns =UnSupport!.
Detection / hold / confirmation setting commands			
sena	sena sena 40	40 / 1 ~ 99 (log scale)	Sets the sensitivity for detecting a target; does not affect targets already detected. The higher the value, the smaller the movements recognized as targets. If too high, the sensor reacts sensitively to surrounding noise.
senb	senb senb 40	40 / 1 ~ 99 (log scale)	Sets the sensitivity for keeping a detected target; does not affect initial target detection. The higher the value, the more readily a detected target is kept even with slight movements. If too high, a target may not disappear because of surrounding noise even after it has actually left.
senin	senin senin 6	2 / 1 ~ 9	Number of consecutive frames required to confirm a target for the first time. The higher the value, the more likely only real targets are confirmed, but confirmation takes longer. It is stored with the same range in all modes (including 3 and 4), but modes 3/4 (presence detection) do not use the coordinate tracker, so this value is not used in the presence decision.
senout	senout senout 5	5 / 2 ~ 50 s (1 s steps) — modes 3 and 4: 20 ~ 500 s (10 s steps), default 300 s / mode 13: default 6 s	Time a detected target (in modes 3/4, an occupied zone) is kept. The higher the value, the longer it is kept. In modes 3/4, entering a value that is not a multiple of 10 returns =OverRange! (it is not rounded). The factory default in mode 13 is 6 s and is linked to the fall SOS decision, so change it with care. On devices where another value is already stored, enter senout 6 once in mode 13 to change it to 6 s.
op	op op 1	1 / 0 ~ 2	Selects the channel combination used for angle estimation. 0: phase method for both left-right and up-down, 1: left-right grid channels + phase (default), 2: up-down grid channels + phase. Stored per mode; unless there is a special reason, use the default value 1.
gnd	gnd gnd 7	7 (mode 2/12/13) / 0 ~ 31	In ceiling mounting modes (2, 4, 12, 13), turns ground-coordinate-based processing functions on/off bit by bit. bit0 (1): ground coordinates for the safe zone, bit1 (2): ground-distance gate for stationary target hold decisions, bit2 (4): ground-distance method for overlapping reflection (shadow) suppression. bit3 and bit4 are reserved (no effect at present). The query response is =GND:m:<applied value> e:<effective value> s:<1 stored/0 default>!. Returns =UnSupport! in wall mounting modes. Changing bit0 resets the targets being tracked. To return to the default, use mreset.
Commands for restroom mode (RESTROOM, mode 12) only			
rstc	rstc rstc 10	5 / 1 ~ 30 (min)	Sets the reference time for the stay-time-exceeded (time over) decision.

rrsh	rrsh rrsh 220	Factory default 250cm (resetting to the actual installation height is strongly recommended) / 100 ~ 300cm	Sets the ceiling sensor installation height. Returns =UnSupport! in other modes.
rrdoorx	rrdoorx rrdoorx -20	0 / Within the current xmin ~ xmax range (cm)	Entrance X coordinate in restroom mode. If it does not match the actual entrance position, exit decisions may be delayed.
rrdoorz	rrdoorz rrdoorz 80	100 / Within the current zmin ~ zmax range (cm)	Entrance Z coordinate in restroom mode. If it does not match the actual entrance position, exit decisions may be delayed.
rrdoorwd	rrdoorwd rrdoorwd 100	80 / 60 ~ 120cm	Diameter of the entrance detection range in restroom mode (radius = value/2). If the person is lost after an approach near the door has been confirmed, the state switches to vacant quickly or with a delay depending on the situation.
Commands for hospital bed mode (HOSPITAL_BED, mode 13) only			
hbsh	hbsh hbsh 220	220 / 200 ~ 300cm (at least hbh + 100cm)	Ceiling sensor installation height. Returns =UnSupport! in other modes.
hbh	hbh hbh 55	55 / 10 ~ 100cm (but no more than hbsh - 100cm)	Bed height (from the floor to the top of the mattress). If set to 30cm or less, the bed is treated as a floor mattress and the fall detection function is turned off automatically.
hbw	hbw hbw 90	90 / 80 ~ 170cm	Bed width. Applied directly to the geometry recalculation.
hboffset	hboffset hboffset 5	5 / 1 ~ 9 (5 = center)	Left-right offset step. 5 is center (0cm), 1 is -20cm, 9 is +20cm.
hbwl hbwr	hbwl hbwr 1	0 / 0, 1	If the left/right side of the bed is completely against a wall, setting 1 blocks the detection area on that side entirely (0 = not blocked). Setting one side to 1 automatically sets the other side to 0; both sides cannot be blocked at the same time. Unlike hbdl/hbdr (curtain height filter), this excludes the detection area itself.
hbdl hbdr	hbdl hbdr 1	0 / 0, 1	Left/right curtain height filter. The detection area is kept, but only reflections higher than the bed (curtains, etc.) are filtered out, while low positions (e.g. a person who has fallen on the floor) are still detected.
hbdlh hbdll hbdrh hbdrl	hbdlh 1	0 / 0, 1	Individually excludes the four bed corners (left-head/left-foot/right-head/right-foot) from detection. Use this when fixed furniture such as a wardrobe at that corner causes false detections; it does not affect detection of the actual patient in the center of the bed. The head/foot detection range itself is always fixed to the inside of the bed, and there is no command to turn it off entirely.
hbfalla	hbfalla 1	1 / 0, 1	Turns the fall-risk warning output on/off.
hbfalls	hbfalls 1	1 / 0, 1	Turns the fall (SOS) notification output on/off.
hbedsorea	hbedsorea 1	1 / 0, 1	Turns the pressure-ulcer risk notification output on/off.
hbedsores	hbedsores 1	1 / 0, 1	Turns the high pressure-ulcer risk (SOS) notification output on/off.
hrv	hrv 1	1 / 0, 1	Turns heart-rate variability (HRV) output on/off. Measurement in a stable posture over a long period is recommended.
Fall detection commands for the wall mounting vital sign and position measurement combined mode (mode			

6)			
fallsh	fallsh fallsh 220	90 / 10 ~ 300cm, 0 = restore default	Sets the sensor installation height used for fall detection. Stored regardless of mode (used mainly in mode 6) and takes effect immediately without a reboot.
falleh	falleh falleh 60	50 / 10 ~ fallsh value, 0 = restore default	Positions higher than this height are excluded from fall decisions. Cannot exceed the value set with fallsh; takes effect immediately without a reboot.
Detection exclusion zone (Exclusion Zone) setting commands			
exzon	exzon exzon 1	0 / 0, 1	Sets or queries the detection exclusion zone function of the currently active mode/install/work profile. 0 disables only the function while keeping the stored slot data. Also works normally in modes 3/4 (presence detection).
exzinfo	exzinfo exzinfo 1	- / slot 1 ~ 5	Queries the number of stored active slots (0~5) or the coordinates (cx, cz) and half-widths (hx, hz) of a specified slot.
exzset	exzset 50 150 exzset 50 150 30 60	slot 1 ~ 5 / hx, hz: 1 ~ 255	Registers coordinates directly in an empty slot of the currently active profile. If hx and hz are omitted, the default of 40cm is applied. Also works normally in modes 3/4 (presence detection).
exzpick	exzpick 3 exzpick 3 30 60	target_num 1 ~ 5 / hx, hz: 1 ~ 255	Reads the coordinates of the currently output target number and registers them automatically in an empty slot. Returns =Missing! if the target number does not exist and =NoTarget! if the specified target does not exist. Modes 3/4 (presence detection) do not use the coordinate tracker and return =UnSupport! (register directly with exzset).
exzdel	exzdel 2 / exzdel 0	slot 0 ~ 5	slot 1~5 deletes an individual slot; 0 deletes all slots of the currently active profile and also turns the function OFF.
Presence Block Exclusion setting commands (modes 3/4 only)			
pblkset	pblkset 120	idx 0 ~ 255	Excludes zone number idx of the 256 zones (blocks) from detection. Zone numbers follow the character order of the occupancy bitmap, with 0 being the first bit on the left, and increase row by row in units of 16 across (idx = distance-direction number x 16 + left-right number). Stored independently for the wall and ceiling profiles with no limit on the number, and zones already confirmed as occupied are cleared immediately. The response is =idx!; =UnSupport! if not in mode 3/4, and =OverRange! if out of range.
pblkdel	pblkdel 120	idx 0 ~ 255	Removes zone idx from the exclusion list. The response is =idx!.
pblkreset	pblkreset	-	Clears all excluded blocks of the current profile. The response is =PblkReset!.
pblkinfo	pblkinfo	-	Queries the excluded blocks of the current profile as a 64-digit hex bitmap (same order as the occupancy bitmap). The response is =PBLK:<64-digit hex>!.

10.1 Initial setup procedure by mode

- The following summarizes the physical installation method and recommended UART command order when installing each mode for the first time. For modes where absolute values for installation height and angle are not specified in the firmware or documentation, it is recommended to install according to site conditions and then fine-tune with the detection area (ROI) commands.

10.2 Mode 1 — Wall position measurement (DISTANCE_WALL) initial setup

- Install fixed on a wall. Absolute values for the recommended installation height and angle are not specified in the firmware/documentation, so install according to site conditions and then fine-tune with the detection area (ROI) commands. If the sensor is tilted, compensate for the downward tilt angle with walltilt (0~45°, default 0 = horizontal); if the cable direction has changed, correct the coordinate axes with rotate (0/90/180/270°).
- 1. Enter mode 1.
  2. If needed, correct the coordinate axes with rotate.
  3. If needed, compensate for the downward tilt angle with walltilt.
  4. Adjust the detection area (ROI) with xmin/xmax/ymin/ymax/zmin/zmax (defaults x -450~450cm, y -100~100cm, z 40~900cm).
  5. Adjust the left-right and up-down detection angles with alrmin/alrmax/adumin/adumax.
  6. Adjust sensitivity and hold time with sena/senb and senin/senout.
  7. If needed, turn on extended output with exten.
  8. After start, check the output packets (start #, vacant #0!).

10.3 Mode 2 — Ceiling position measurement (DISTANCE_CEILING) initial setup

- Install fixed on the ceiling, facing downward. Absolute values for the recommended installation height and angle are not specified in the firmware/documentation. Because it is ceiling-mounted, ymax cannot exceed 0cm (no positive Y).
- 1. Enter mode 2.
  2. If needed, correct the coordinate axes with rotate.
  3. Adjust the ceiling ROI with xmin/xmax/ymin/ymax/zmin/zmax (defaults x -450~450cm, y -450~-40cm, z -450~450cm).
  4. Adjust the detection angles with alrmin/alrmax/adumin/adumax.
  5. Adjust sensitivity with sena/senb and senin/senout.
  6. After start, check the output packets (start $, vacant $0!).

10.4 Mode 3 — Wall presence detection (PRESENCE_WALL) initial setup

- Install fixed on a wall. Instead of tracking individual coordinates, this mode detects by zone (block), dividing 8m x 8m into 256 zones of 50cm. Absolute values for the recommended installation height and angle are not specified.
- 1. Enter mode 3.
  2. If needed, correct the coordinate axes with rotate (walltilt returns =UnSupport! in this mode).
  3. Adjust the detection area with xmin/xmax/ymin/ymax/zmin/zmax and alrmin/alrmax/adumin/adumax (defaults x ±400cm, y ±200cm, z 0~800cm, angle ±75°).
  4. Adjust detection sensitivity and grid channels with sena/senb/op (senin is not used in the presence decision of this mode).
  5. Set the zone occupancy hold time with senout in the range 20~500 s (10 s steps, default 300 s).
  6. If needed, register fixed reflectors such as fans and furniture as exclusion zones with exzon/exzset.
  7. After start, check the output packets (start *, vacant *0!).

10.5 Mode 4 — Ceiling presence detection (PRESENCE_CEILING) initial setup

- Install fixed on the ceiling, facing downward. The decision method is the same as mode 3; only the

roles of the coordinate axes differ (X = left-right, Z = the other horizontal axis, Y = height from the ceiling toward the floor).

- 1. Enter mode 4.
  2. If needed, correct the coordinate axes with rotate.
  3. Adjust the ceiling ROI with xmin/xmax/zmin/zmax (defaults x ±400cm, z ±400cm; the command itself accepts up to ±450cm). ymin/ymax default to -300~-30cm; adjust the detection angles with alrmin/alrmax/adumin/adumax.
  4. Adjust detection sensitivity and grid channels with sena/senb/op.
  5. Set the zone occupancy hold time with senout in the range 20~500 s (10 s steps, default 300 s).
  6. If needed, register fixed reflectors as exclusion zones with exzon/exzset.
  7. After start, check the output packets (start ~, vacant ~0!).

10.6 Mode 5 — Wall vital sign (VITAL) initial setup

- Install fixed on a wall. Absolute values for the recommended installation height and angle are not specified; the downward tilt angle can be compensated with walltilt (0~45°). Because the nearest movement strongly affects the result, install so that no other people or objects come between the sensor and the subject.
- 1. Enter mode 5.
  2. Correct the physical orientation with walltilt/rotate.
  3. Narrow the ROI to the subject's position with xmin/xmax/ymin/ymax/zmin/zmax (defaults x -125~125cm, y -150~100cm, z 40~250cm).
  4. Set the heartbeat and respiration distance limits with hmax/bmax.
  5. If needed, turn on waveform output with wave 1 (automatically fixed at 10Hz) and extended output with exten 1.
  6. Adjust sensitivity with sena/senb and senin/senout.
  7. After start, check the output packets (start %, or start @ with wave 1).
- Apnea is not a separate status code but appears as a respiration rate close to 00, so judge it together with the waveform.

10.7 Mode 6 — Wall vital sign + position (VITAL_POSITION) initial setup

- Install fixed on a wall. Absolute values for the recommended installation height and angle are not specified; the downward tilt angle can be compensated with walltilt.
- 1. Enter mode 6.
  2. Correct the physical orientation with walltilt/rotate.
  3. Adjust the ROI with xmin/xmax/ymin/ymax/zmin/zmax (defaults x -400~400cm, y -100~100cm, z 40~630cm).
  4. Set the heartbeat and respiration distance limits with hmax/bmax.
  5. If needed, set wave 1 (exten does not affect the output of this mode).
  6. After start, check the output packets (start &, or start ? with wave 1), and confirm that vital-sign data is valid only in the first slot.
  7. If needed, adjust the sensor installation height and exclusion height used for fall decisions with fallsh/falleh (takes effect immediately without a reboot).

10.8 Mode 11 — Advanced vital sign (ADV_VITAL) initial setup

- The default geometry is similar to mode 5, but the y-axis range differs (x ±125cm, y -100~100cm, z 40~250cm). Absolute values for the recommended installation height and angle are not specified.
- 1. Enter mode 11.
  2. Correct the physical orientation with walltilt/rotate.
  3. Adjust the ROI with xmin/xmax/ymin/ymax/zmin/zmax (defaults x ±125cm, y -100~100cm, z 40~250cm — the y-axis range differs from mode 5).
  4. Set hmax/bmax.

5. Measure in a stable posture for a sufficient time (several minutes or more) so that HRV indicators such as RMSSD/SDNN converge stably.
6. If needed, set wave 1 and exten 1.
7. After start, check the output packets (start }, or start { with wave 1).

- No additional configuration commands specific to mode 11 were identified in the firmware; it basically uses the same command set as mode 5.

10.9 Mode 12 — Restroom ceiling (RESTROOM) initial setup

- Install fixed on the ceiling. The installation height must be reset to the actual installation height with the rrsh command (100~300cm, factory default 250cm); this command cannot be used in other modes (=UnSupport!).
- 1. Enter mode 12.
  2. Set the actual ceiling installation height with rrsh in the range 100~300cm (set this first, because the geometry is calculated from this value).
  3. Adjust the ROI to the narrow restroom stall with xmin/xmax/zmin/zmax and alrmin/alrmax/adumin/adumax (defaults x -50~50cm, z -40~100cm, angle -25~25°).
  4. Set the stay time limit (1~30 min) with rstc.
  5. Set the actual entrance coordinates with rrdoorx/rrdoorz and the entrance detection diameter (60~120cm) with rrdoorwd (these are key to exit decisions through the door, so always use measured values).
  6. If needed, set wave 1 (continuous output only while sitting/lying and not in an alarm state).
  7. After start, check the output packets (start ^, vacant ^0!).
- The exten command does not affect the output of this mode. If the person is lost for about 0.8 s or more after an approach near the door has been confirmed, the state switches to vacant immediately; other losses (e.g. a momentary loss of detection while the person stays still) keep the last state for 10 minutes before switching to vacant. If the entrance position (rrdoorx, rrdoorz, rrdoorwd) is not set to match the actual door, every exit will depend solely on this 10-minute wait.

10.10 Mode 13 — Hospital bed ceiling mounting initial setup

- When the bed touches the wall with the head toward the wall, install the sensor about 75cm from the wall. Align the sensor with the center of the bed in the left-right direction (within ±20cm), and if the bed position shifts left or right after installation, compensate with hboffset.
- 1. Enter mode 13.
  2. Set the sensor installation height with hbsh in the range 200~300cm (default 220cm).
  3. Set hbh (bed height 10~100cm, but no more than hbsh - 100cm) and hbw (bed width 80~170cm).
  4. If one side of the bed is against a wall, block detection on that side with hbwl/hbwr (0/1 each); reduce false detections caused by curtains with hbdl/hbdr (0/1 each, curtain height filter).
  5. If fixed furniture such as a wardrobe at a bed corner causes false detections, exclude only that corner (head/foot × left/right) with hbdlh/hbdll/hbdrh/hbdrl (0/1 each).
  6. If needed, correct the left-right offset with hboffset (1~9, 5 = center).
  7. Set display options with hbfalla/hbfalls, hbedsorea/hbedsores and hrv.
  8. After start, check the status packets (start /, or start ₩ with wave 1).

11 Order Information

AS_613_P/C/U		
AS_613	P	Installing 2.54mm pin header
	C	Installing connector
	U	Installing USB-C connector
ex) AS_613_U		Model with USB-C connector only installed
ex) AS_613_PU		Model with 2.54mm pin header and USB-C connector installed
ex) AS_613_CU		Model with connector and USB-C connector installed

Warning

This device operates wirelessly and may emit radio waves that can interfere with the proper operation of pacemakers or implanted medical devices. Do not use the device within 20cm of a pacemaker or other implanted medical device. If you use a pacemaker or other implanted medical device, do not use this device without first consulting your doctor or the manufacturer of the pacemaker or implanted medical device.

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